



Cell2location maps fine-grained cell types in spatial transcriptomics

Vitalii Kleshchevnikov¹, Artem Shmatko^{1,2}, Emma Dann¹, Alexander Aivazidis¹, Hamish W. King^{1,3}, Tong Li¹, Rasa Elmentaite¹, Artem Lomakin^{1,4}, Veronika Kedlian¹, Adam Gayoso^{1,5}, Mika Sarkin Jain^{1,7}, Jun Sung Park^{1,4}, Lauma Ramona¹, Elizabeth Tuck¹, Anna Arutyunyan¹, Roser Vento-Tormo¹, Moritz Gerstung^{1,4,5}, Louisa James^{1,3}, Oliver Stegle^{1,5,8}✉ and Omer Ali Bayraktar¹✉

Spatial transcriptomic technologies promise to resolve cellular wiring diagrams of tissues in health and disease, but comprehensive mapping of cell types in situ remains a challenge. Here we present cell2location, a Bayesian model that can resolve fine-grained cell types in spatial transcriptomic data and create comprehensive cellular maps of diverse tissues. Cell2location accounts for technical sources of variation and borrows statistical strength across locations, thereby enabling the integration of single-cell and spatial transcriptomics with higher sensitivity and resolution than existing tools. We assessed cell2location in three different tissues and show improved mapping of fine-grained cell types. In the mouse brain, we discovered fine regional astrocyte subtypes across the thalamus and hypothalamus. In the human lymph node, we spatially mapped a rare pre-germinal center B cell population. In the human gut, we resolved fine immune cell populations in lymphoid follicles. Collectively, our results present cell2location as a versatile analysis tool for mapping tissue architectures in a comprehensive manner.

The cellular architecture of tissues, where distinct cell types are organized in space, underlies cell–cell communication, organ function and pathology. Emerging spatial transcriptomics technologies provide key opportunities to map resident cell types and cell signaling in situ in a scalable manner. Despite existing proof-of-concept applications of these technologies^{1–6}, it remains a challenge to define spatial transcriptomics workflows that allow comprehensive mapping of resident cell types in tissues. One reason is the enormous cellular diversity across organs, including myriad fine-grained cell types such as immune^{7,8}, stromal^{7,9} and neuronal^{10,11} subpopulations. These fine cell types, marked by subtle transcription differences, are often specialized across tissues and yield large potential for discoveries. Another reason is the diversity of spatial tissue architecture, ranging from the brain featuring discrete anatomical regions with distinct cell types to immune organs featuring dynamically modified microenvironments with fine-grained cell types. To create comprehensive cellular wiring diagrams of tissues with spatial transcriptomics, it is necessary to establish experimental and computational methods that are sufficiently sensitive to resolve fine-grained differences in cell types across diverse tissues.

Strategies that generate coupled single-cell and spatially resolved transcriptomics offer a scalable approach to address these challenges. The key principle is to first identify resident cell types based on single-cell RNA sequencing (scRNA-seq) from dissociated tissues and then map the identified cell types to their tissue positions in situ based on spatial transcriptomic profiles¹². Here, spatial RNA-seq technologies, such as Visium¹³, HDST¹⁴ and Slide

sequencing (Slide-seq)², where mRNAs are positionally captured from thin tissue sections using microarray or bead array grids, provide transcriptome-wide data at high throughput and rely on simple histology and molecular biology protocols.

A key limitation of existing grid-based profiling methods is the lack of single-cell resolution, as spatial RNA-seq measurements combine multiple cells (Visium and Tomo-Seq)^{13,15} or fractions of multiple cells (Slide-seq and HDST)^{2,14} at each tissue location. This challenge can be overcome by the computational integration of spatial RNA-seq with the reference transcriptome signatures of cell types obtained from coupled scRNA-seq profiles. Although, recently, several computational methods have been proposed for this task^{1–3,5,6}, it is not clear whether these methods offer sufficient sensitivity to resolve fine-grained cell types in complex tissues, which is critical to make full use of the discovery potential of spatial transcriptomic data. Additionally, different confounding sources of variation need to be accounted for, which enables more complex integration tasks of data that consist of multiple experiments, including variable cell numbers across tissue positions and variable total mRNA content across cell types and individual cells^{16,17}.

To address the aforementioned challenges, we present cell2location, a Bayesian model that is designed to resolve fine-grained cell types in spatial transcriptomic data and create cellular maps of diverse tissues. Cell2location can handle complex experimental settings, performing joint analysis of multiple scRNA-seq and spatial transcriptomic datasets. Unique to the model is a hierarchical design to borrow statistical strength across locations, which yields

¹Wellcome Sanger Institute, Hinxton, Cambridge, UK. ²Moscow State University, Leninskie Gory, Moscow, Russia. ³Centre for Immunobiology, Blizard Institute, Queen Mary University of London, London, UK. ⁴European Molecular Biology Laboratory, European Bioinformatics Institute (EMBL-EBI), Hinxton, UK. ⁵European Molecular Biology Laboratory, Genome Biology Unit, Heidelberg, Germany. ⁶Center for Computational Biology, University of California, Berkeley, Berkeley CA, USA. ⁷Theory of Condensed Matter, Department of Physics, Cavendish Laboratory, University of Cambridge, Cambridge, UK.

⁸Division of Computational Genomics and Systems Genetics, German Cancer Research Center (DKFZ), Heidelberg, Germany.

✉e-mail: oliver.stegle@embl.de; ob5@sanger.ac.uk

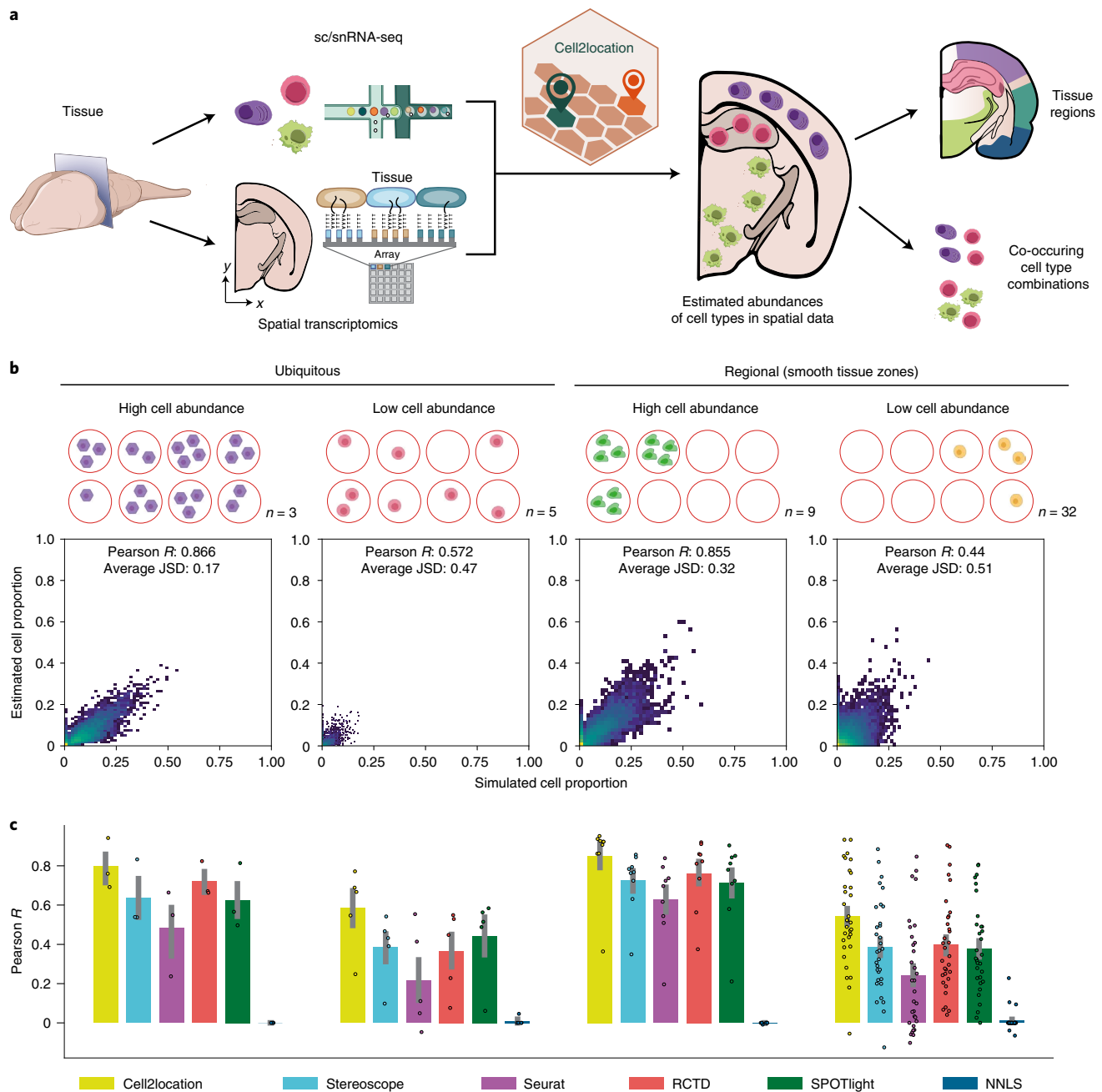


Fig. 1 | Cell2location overview and validation using simulated data. a, Overview of the spatial mapping approach and the analysis workflow of cell2location. From left to right: scRNA-seq and spatial transcriptomics profiles are generated from the same or related tissues. Cell2location takes reference cell type signatures derived from scRNA-seq and spatial transcriptomics data as input. The model then decomposes spatially resolved multi-cell RNA count matrices into the reference cell type signatures, thereby estimating cell abundance of individual cell types across locations. **b**, Model validation using simulated data. A benchmark dataset is constructed by combining cells drawn from 49 reference cell types (obtained from mouse brain scRNA-seq; Fig. 2b), using alternative cell type abundance patterns. Top: four alternative cell type abundance patterns considered for data simulation (see also Supplementary Fig. 2) and the number of cell types assigned to each pattern (Methods). Bottom: 2D histogram plots, displaying the concordance between simulated (x axis) and estimated (y axis) cell type proportions across 2,500 locations, aggregated for cell types assigned to the respective pattern. Color denotes 2D histogram count (50 bins along both the x axis and y axis). Pearson R denotes Pearson correlation, and JSD denotes Jensen-Shannon divergence. **c**, Assessment of cell2location and alternative methods for estimation of cell type proportions for 49 cell types from the four respective abundance patterns (columns, x axis, 2,500 locations and 49 cell types). Shown are bar plots of Pearson correlation coefficients (y axis, R) between simulated and estimated cell type proportions for individual cell types across locations, considering alternative methods (color, x axis). Bar height denotes the average correlation coefficient across cell types; error bars denote standard errors.

improved resolution and sensitivity, in particular for resolving fine-grained cell types in complex tissues. Here we assess cell2location in three different tissues, where we consistently show improved

sensitivity for mapping fine-grained cell types. Moreover, we map rare B cell states in the human lymph nodes and identify regional astrocyte subtypes in the mouse brain.

Results

Cell2location: a Bayesian model for spatial mapping of cell types.

Cell2location maps the spatial distribution of cell types by integrating single-cell or nuclei RNA-seq and spatial transcriptomic data from a given tissue (Fig. 1a). The first step of the cell2location workflow is to estimate reference cell type signatures from single-cell and single-nucleus RNA-seq profiles, which correspond to gene expression profiles for a set of user-provided cell types and subpopulations identified using conventional clustering or other methods. By default, cell2location uses negative binomial regression for this step, which allows robust combination of data across technologies and batches (Methods and Supplementary Fig. 1). In a second step, cell2location uses these reference signatures and one or multiple spatial transcriptomic datasets as input to decompose mRNA counts at individual spatial locations into reference cell types (Fig. 1a and Supplementary Fig. 1).

Although cell2location is related to existing transcriptomic decomposition methods^{1,3,5,6,18}, it offers features to (1) borrow statistical strength across locations with similar cell composition, (2) account for batch variation across slides as well as variation in mRNA detection sensitivity and (3) estimate absolute cell type abundances by incorporating prior information about the analyzed tissues (see Supplementary Methods for details). In addition, cell2location is computationally efficient, owing to variational approximate inference and GPU acceleration. The cell2location software is integrated with the scvi-tools framework¹⁹ and comes with a suite of downstream analysis tools, including the identification of groups of co-located cell types.

To validate cell2location, we initially considered simulated data that were generated to reflect diverse cell abundance patterns across ubiquitous and spatially restricted cell types (Fig. 1b). In brief, we created a spatial transcriptomics dataset with 2,500 locations by sampling cells from 49 reference cell types, using a combination of ubiquitous abundance patterns and cell types distributed according to smooth ‘tissue zones’ as observed in real data (Fig. 1b, Supplementary Fig. 2 and Methods). Reference cell type signatures and other simulation parameters were adapted from a mouse brain scRNA-seq dataset (Methods). Of note, unlike most benchmark datasets on real data (see below), this synthetic dataset provides ground truth information on the presence and absence of cell types as well as on quantitative abundance.

We then used the cell2location to estimate cell type proportions for different cell types at individual locations. Comparison of these estimates to the simulated (true) cell type proportions revealed that cell2location mapped cell types with high accuracy, irrespective of

the cell abundance pattern (Fig. 1b), including low abundant and regional cell types.

Next, we compared cell2location to recently proposed methods for the estimation of relative cell type abundance from spatial transcriptomics data (Stereoscope³, Seurat¹, RCTD⁵, NNLS (AutoGeneS)^{18,20} and SpotLIGHT⁶). Across the considered cell abundance patterns, we found that cell2location yielded more accurate estimates of the cell type proportions across locations than other methods (Fig. 1c). In addition to this quantitative assessment of cell type proportions, we also assessed the ability of alternative methods to determine locations where individual cell types are present—that is, have non-zero cell abundance—which yielded a consistent ranking of methods (Supplementary Fig. 3b).

We note that, in addition to estimating relative cell type proportions, cell2location can incorporate prior information about the tissue to estimate absolute cell type abundance, which, again, were highly concordant with the simulated ground truth (Supplementary Fig. 4a). Finally, we explored how specific design choices of cell2location contribute to its performance (Supplementary Fig. 4b), and we assessed the robustness of the model to incomplete reference cell type sets (Supplementary Fig. 4c) as well as variation in hyper-parameter settings (Supplementary Fig. 5).

Cell2location accurately maps mouse brain cell types. To validate cell2location in a real tissue, we first applied the model to data from the mouse brain, which features diverse neural cell types organized in a well-characterized spatial architecture across brain areas, thus presenting a canonical use case to test spatial genomics^{1,3,5,6}. We generated matched single nucleus (sn) and Visium spatial RNA-seq (10x Genomics) profiles of adjacent mouse brain sections that contain multiple regions from the telencephalon and diencephalon (Fig. 2a). To assess the biological and intra-organ technical variation in spatial mapping, we assayed two mouse brains and serial tissue sections from each brain (total of three and two matched sections from two animals, respectively, and an extra section for snRNA-seq), creating a rich multi-modal and replicated transcriptomic dataset.

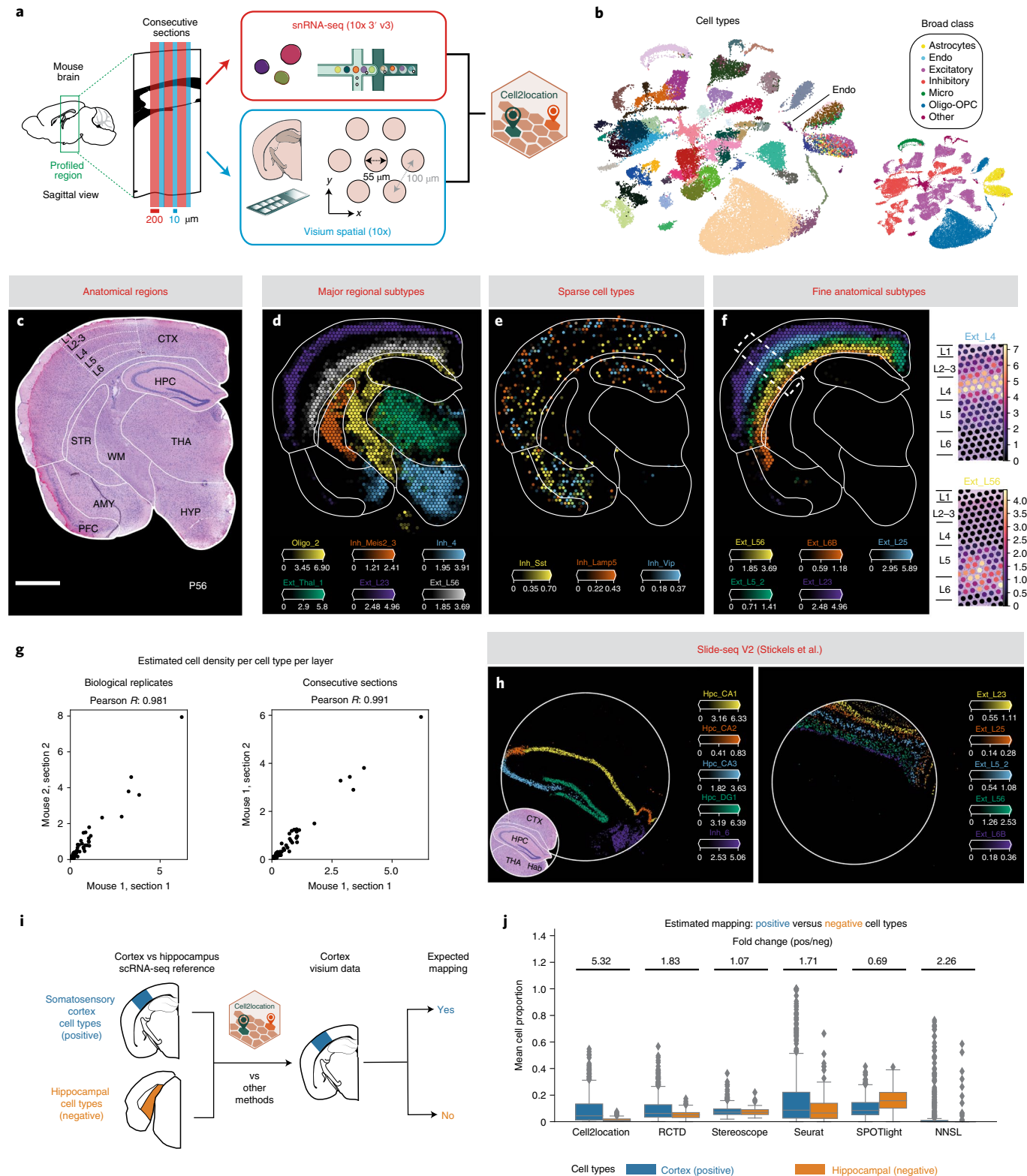
To define reference cell type signatures, we pooled snRNA-seq data across replicate animals and tissue sections (40,532 nuclei) and applied a conventional scRNA-seq workflow followed by Louvain clustering (Methods). This identified 51 putative cell type clusters, which were well represented across replicates (Supplementary Fig. 6). Of these, 44 clusters could be identified as expected cell types in the brain (Fig. 2b, Supplementary Fig. 7 and Methods), including astrocytes, excitatory or inhibitory neurons or glia subtypes, annotated by markers from prior literature^{10,16,21}. The two astrocyte clusters

Fig. 2 | Comprehensive and accurate spatial mapping of cell types in the mouse brain. **a**, Experimental workflow. Adjacent tissue sections from two mice were used to generate single-nucleus and spatial data, across a total of five pairings of Visium and snRNA-seq, respectively (and an extra section for snRNA-seq). **b**, UMAP representation (x and y axis) of broad cell type classes (right) and 59 cell subtypes (left; see Supplementary Fig. 6a for labels), as identified by Louvain clustering of the integrated snRNA-seq dataset across sections from **a**. **c**, Mouse brain regions present in the assayed section. H&E image of the tissue section and the outlines of brain regions are shown. Scale bar, 1 mm. **d**, Estimated cell abundances (color intensity) of major cell types (color) across regions. **e**, Estimated cell abundances (color intensity) of sparse inhibitory neurons (color). **f**, Estimated cell abundances (color intensity) of cortical excitatory neurons (color). Small panels show the estimated cell abundance (color) for two select subtypes on the region indicated with the white bounding box. **g**, Consistency of cell2location cell abundance estimates across replicate slides (x and y axis), considering either replicate animals (left) or consecutive serial sections from the same animal (right). Considered are cell abundance estimates across 59 reference cell types across cortical layers; *R* denotes Pearson correlation. **h**, Spatial mapping in the Slide-seq V2 data. Left, estimated cell abundances (color intensity) of hippocampal neuron types (Hpc subtypes) and a thalamic habenular neuron type (Inh_6). Right, estimated cell abundances (color intensity) of cortical excitatory neurons. **i**, Benchmarking workflow for assessing the spatial mapping specificity of cell2location and alternative methods. scRNA-seq reference cell types¹¹ from manually dissected SSp cortex layers (top, positive set) and the par-/post-/pre-subiculum divisions of the hippocampus (bottom, negative set) were mapped to Visium data from the SSp cortex. **j**, Distribution of estimated cell type proportions at locations with high abundance of each cell type in the spatial data for positive cell types (SSp cortex, orange, x axis, *n* = 98) and negative cell types (hippocampus, blue, x axis, *n* = 23) in the SSp cortex region for alternative methods. Box plots show the distribution of cell proportions (99.8% quantile across locations within the SSp cortex) for each cell type and each replicate section (Supplementary Fig. 11e; *n* = 5 sections). Fold changes denote the difference in average 99.8% quantile of cell type proportions between positive and negative cell types. Box center, bounds and whiskers show median, quartiles and data points that lie within 1.5x interquartile range of the lower and upper quartiles, respectively. The observations that fall outside this range are displayed as dots.

could be further subdivided into ten transcriptional subtypes (see next section). Collectively, this resulted in reference cell type signatures for 59 distinct and annotated cell subtypes, which comprehensively cover the profiled brain areas (Supplementary Fig. 7).

We applied cell2location to map these cell types to spatial locations of the mouse brain Visium data, across replicate animals and sections. The resulting cell type mapping was markedly consistent

with anatomical locations expected from prior literature (Fig. 2c–f and Supplementary Files 1–6). Our results captured diverse cell type patterns, resolving broad regional cell types such as excitatory neurons of thalamus and oligodendrocytes of white matter (Fig. 2d)^{10,22}, rare subtypes such as cortical interneurons (Fig. 2e)^{23,24} and sub-regional subtypes such as excitatory neurons across cortical layers (Fig. 2f and Supplementary Fig. 8e)^{23,24}. The cell type



abundance estimates were highly concordant across adjacent tissue sections and replicate animals, as quantified across cortical layers (Fig. 2g and Supplementary Fig. 9). Finally, we clustered tissue locations based on their estimated cell type abundance profiles (Methods), which yielded clusters that recapitulate the anatomical brain areas, with the reference cell types mapping to the expected tissue regions (Supplementary Fig. 10).

Next, to assess the specificity of cell2location and alternative methods, we mapped a large existing scRNA-seq dataset that includes cell types from multiple brain regions¹¹, only some of which are present in our spatial data (Fig. 2i). Consequently, this dataset spans ‘positive’ cell types from the somatosensory (SSp) cortex ($n = 98$) as well as ‘negative’ cell types that are exclusively present in the hippocampus ($n = 23$)¹¹. When mapping these cell types to our SSp cortex Visium data, we would expect that negative cell types are excluded and only positive cell types are assigned substantial cell type proportions.

Indeed, cell2location mapping yielded substantially lower cell proportion estimates for the negative compared to the positive cell types (Fig. 2j; positive/negative fold change = 5.1). By contrast, other methods yielded a substantially weaker contrast between negative and positive cell types (Fig. 2j; positive/negative fold change = 0.7–1.9 for other methods; Supplementary Fig. 11). This indicates that cell2location accurately excludes reference cell types that should not be present in a spatial dataset, whereas other methods tend to ‘overfit’ reference cell types. We also used cell type proportion estimates derived from the scRNA-seq data, which were generated from manually dissected cortical layers, to confirm the quantitative cell type proportions estimated by cell2location (Supplementary Fig. 11 and Supplementary File 7).

Finally, to show the versatility of cell2location across spatial transcriptomic technologies, we mapped our mouse brain snRNA-seq reference to a Slide-seq dataset of the mouse brain that offers 10- μ m spatial resolution²⁵. Cell2location resolved cell types in Slide-seq spatial data, including fine anatomical subtypes across hippocampal divisions and cortical layers (Fig. 2h, Supplementary Fig. 12a and Supplementary Files 8 and 9). Moreover, cell2location revealed that more than 50% of Slide-seq ‘beads’ (that is, 10- μ m-diameter spatial locations) contained more than one cell type (Supplementary Fig. 12b), consistent with previous observations⁵.

Collectively, our results show that cell2location maps diverse brain cell types with high accuracy and reproducibility and that the method is broadly applicable to spatial transcriptomic technologies.

Identification of regional astrocyte subtypes. Next, we assessed cell2location for mapping fine-grained cell types in the mouse brain. We focused on the heterogeneity of astrocytes, which is the largest class of glial cells that intimately regulate neuronal circuit development and function²⁶. Although astrocytes were historically considered to be a homogeneous cell type, recent evidence suggests that astrocytes are regionally diversified across the brain²⁶, including fine anatomical divisions such as cortical layers^{27–29}. However, the extent of the regional heterogeneity of astrocytes is largely unknown, as existing studies have primarily compared their

prevalence across selected brain areas^{30,31} or described broad astrocyte subtypes across major brain divisions (for example, telencephalon versus diencephalon)^{10,32}.

To identify regionally enriched astrocytes, we annotated and spatially mapped ten molecularly and spatially distinct astrocyte subtypes that were identified based on our snRNA-seq reference. In brief, we considered 3,013 cells annotated as astrocytes in our initial analysis (Fig. 2b). We then applied BBKNN³³ to further refine the integration of these cells across all six snRNA-seq datasets (Methods), followed by Leiden clustering, which identified 17 transcriptomically defined subclusters (Supplementary Fig. 13). Next, we performed an initial mapping of these putative cell subtypes using cell2location, merging clusters that were not sufficiently distinct in both their mapped spatial locations and marker gene expression profiles (Supplementary Figs. 13 and 14 and Methods). This resulted in ten molecularly and spatially distinct astrocyte subtypes (Fig. 3a,b and Supplementary Fig. 15), which we considered for further analysis. The extent of molecular heterogeneity across these subtypes (measured by the number of marker genes) was substantially lower than across neuronal subpopulations, confirming the transcriptionally fine-grained nature of these cell populations (Supplementary Fig. 13d).

We identified gray matter astrocyte subtypes that were spatially enriched in the thalamus (THAL), hypothalamus (HYPO), cortex (CTX), amygdala (AMY), hippocampus (HPC) and striatum (STR) as well as white matter astrocytes (Fig. 3b and Supplementary Fig. 15). These subtypes expressed expected regional astrocyte markers¹⁰, such as *Foxg1* in the telencephalon (CTX, AMY, HPC and STR) and *Slc6a11* and *Agt* in the diencephalon (THAL and HYPO) (Supplementary Fig. 15). Moreover, they stratified to finer regional subtypes based on the expression of subtype-specific genes such as *Angpt1* and combinatorially expressed markers such as *Gria1* (Supplementary Fig. 15; see below).

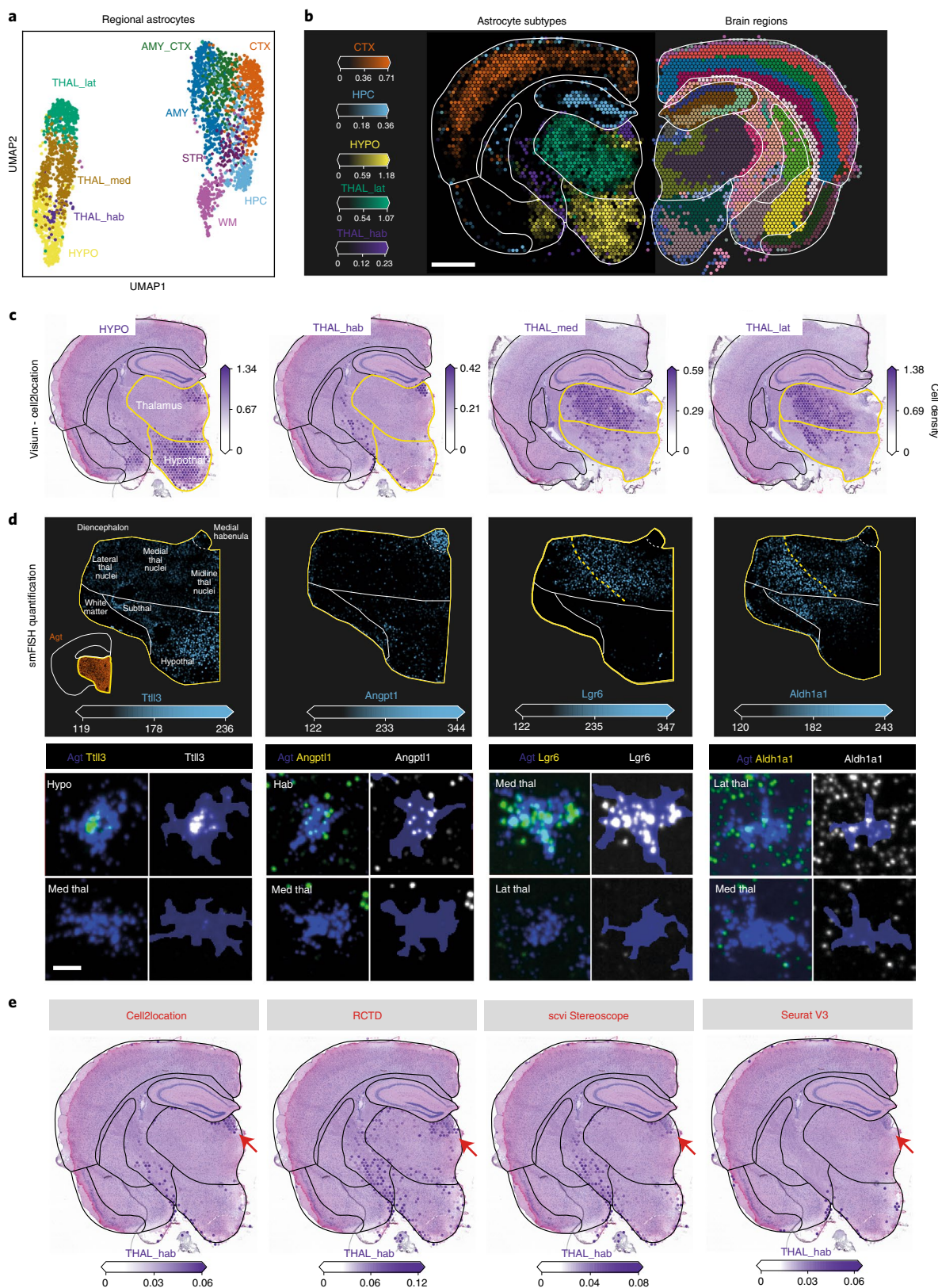
We focused on diencephalic astrocyte subtypes, as only one type of gray matter astrocyte subtype has been described in this brain area to date¹⁰. Notably, our data revealed four regionally distinct gray matter astrocyte subpopulations across the diencephalon (Fig. 3c and Supplementary Fig. 15). HYPO astrocytes were mainly located in the hypothalamus and, to a lesser extent, in the midline thalamic nuclei and the lateral habenula, and they were marked by *Till3* expression. Three THAL subtypes occupied different regions of the thalamus. THAL_hab astrocytes, a rare cell population in our reference (Supplementary Fig. 6d), were restricted to the medial habenula and ventral white matter tracts and specifically expressed *Angpt1*. THAL_med astrocytes were enriched in the medial thalamic nuclei and were marked by *Lgr6* expression. Finally, THAL_lat astrocytes, although present throughout the thalamus and the subthalamus, were most prevalent in ventrolateral thalamic nuclei and were marked by elevated *Aldh1a1* expression compared to other subtypes (Fig. 3c and Supplementary Fig. 15).

We validated the identity and distribution of diencephalic astrocyte subtypes using single-molecule fluorescence in situ hybridization (smFISH)²⁷. We used *Agt* smFISH to specifically label and

Fig. 3 | Spatial discovery of regional astrocyte subtypes. **a**, UMAP representation (x and y axis) of ten molecularly and spatially distinct astrocyte subtypes (color) identified in our snRNA-seq reference data. **b**, Estimated cell abundances (color intensity) of selected regional astrocyte subtypes (left hemisphere) and brain regions as identified using clustering of locations (right hemisphere). Scale bar, 1 mm. **c**, Estimated cell abundances (color intensity) of four diencephalic astrocyte subtypes. Thal_med and Thal_lat are shown from a more representative section from the same mouse brain. **d**, smFISH validation. Top panels: Small map shows locations of diencephalic astrocytes segmented based on *Agt* smFISH. Large maps show quantified subtype marker expression (color intensity) at a single-astrocyte level (dots). Yellow dashed line indicates the medial boundary of high *Aldh1a1* expression. Bottom panels, close-up images of diencephalic astrocyte subtypes. First columns show smFISH for astrocyte marker *Agt* and subtype markers. Second columns show astrocyte segmentation masks (blue) transparently overlaid on subtype genes. $n = 1$ mouse and $n = 3$ technical replicate tissue sections for each marker independently imaged. Scale bar, 10 μ m. **e**, Estimated cell proportions (color intensity) for mapping THAL_hab astrocyte subtypes, considering the four best-performing methods (columns). Red arrows indicate the location of the medial habenula.

segment astrocytes throughout the diencephalon¹⁰ (Supplementary Figs. 16 and 17) and then quantified the expression of co-stained subtype markers at single-astrocyte resolution (Fig. 3d). The in situ expression patterns of regional astrocyte markers (*Till3*, *Angpt1*, *Lgr6* and *Aldh1a1*) validated the spatial mapping of their respective

astrocyte subtypes by cell2location (Fig. 3d). Finally, we also considered alternative tools for spatially mapping regional astrocytes, finding that cell2location was the only transcriptomic decomposition method that accurately mapped THAL_hab astrocytes to their validated tissue location (Fig. 3e and Supplementary Fig. 18).



These results validate diencephalic astrocyte subtypes and show that cell2location can spatially map fine-grained cell types.

Resolving the cellular compartments of the human lymph node.

To test cell2location on diverse tissue architectures, we applied the model to spatially map tissue microenvironments of the human lymph node. Unlike the mouse brain, the human lymph node is characterized by dynamic microenvironments with many spatially interlaced cell populations. We analyzed a publicly available Visium dataset of the human lymph node from 10x Genomics³⁴ and spatially mapped a comprehensive atlas of 34 reference cell types derived by integration of scRNA-seq datasets from human secondary lymphoid organs composed of 73,260 cells^{35–37} (Fig. 4a,b, Supplementary Fig. 19 and Methods).

Histological examination of the lymph node Visium sample revealed multiple germinal centers (GCs). Correspondingly, after mapping with cell2location, the expected major regions within the lymph node tissue could be readily identified, including T cell and B cell zones and GCs with follicular dendritic cells (FDCs) (Fig. 4c, right panel, and Supplementary Files 10 and 11).

Next, we used the cell2location mapping results to identify the spatial co-occurrence of cell types to better understand the tissue organization and predict cellular interactions. We performed non-negative matrix factorization (NMF) of the cell type abundance estimates from cell2location (Fig. 4d). Similarly to the established benefits of applying NMF to conventional scRNA-seq^{38,39}, the additive NMF decomposition yielded a grouping of spatial cell type abundance profiles into components that capture co-localized cell types. This NMF-based decomposition naturally accounts for the fact that multiple cell types and microenvironments can co-exist at the same Visium locations (Supplementary Fig. 20) while sharing information across tissue areas (for example, individual GCs).

We observed that the groups of cell types identified by NMF corresponded to known functionally relevant cellular compartments of the lymph node (Fig. 4d). Furthermore, the NMF output enabled the dissection of the spatial compartments of B cell maturation. Specifically, naive B cells were mapped to a B follicle zone distinct from GCs (Fig. 4e, panel 1). We also identified the GC dark zone where B cells clonally expand and proliferate (Fig. 4e, panel 2) as well as the GC light zone where they undergo selection by T follicular helper cells and FDCs to differentiate into antibody-producing plasmablasts (Fig. 4e, panel 3).

In addition to confirming known biology, the cell2location combined with the NMF decomposition enabled the fine-grained spatial mapping of a rare secondary activation B cell state, which was recently reported as primed to undergo class switch recombination preceding the GC entry^{37,40}. Our data indicate that this population (termed here ‘preGC’) occupies a different tissue

compartment from the upstream activated cells and downstream GC cells (Fig. 4e, panel 4), suggesting that it is more likely to initiate new GC reactions rather than to join existing ones. Finally, our analysis was capable of distinguishing blood vessel zones containing endothelial and vascular smooth muscle cells (Supplementary Fig. 21a, panel 5) from the adjacent perivascular zone enriched in T helper and B memory cells (Supplementary Fig. 21a, panel 6).

Taken together, our results show that cell2location can map complex tissues characterized by spatially interlaced cell types and aid biological interpretation by identifying putatively interacting cell types.

Benchmarking cell2location for mapping fine immune cell types.

Cell atlasing studies have greatly expanded the catalog of fine-grained cell types across human tissues, which are often of key interest for understanding tissue function and pathology^{7,9,11}. To quantitatively benchmark cell2location and alternative methods for mapping transcriptionally fine-grained cell types, we considered immune cell populations across two human tissues where many fine subtypes with specialized immune functions can be identified^{7–9}. For example, T follicular helper cells differ from the classical CD4⁺ T cells in their ability to enter the GCs and guide B cell differentiation into plasma cells and memory B cells to ensure long-lasting immunity⁴¹.

Initially, we examined GC-specific immune cell types in the human lymph node (Fig. 4). We classified six known GC cell types as either transcriptionally fine or transcriptionally distinct, based on the Euclidean distance of their expression signatures to the closest other cell type (Supplementary Fig. 22a and Methods). We annotated the anatomical locations of GCs using Visium histology images (Fig. 4f) and assessed the accuracy of alternative methods in mapping the six GC cell types to the expected GC locations. Although all considered methods achieved high accuracy for mapping distinct cell types such as FDCs (Fig. 4h), cell2location was markedly more accurate in mapping fine-grained cell types such as GC T follicular helper cells (Fig. 4f–h and Supplementary Figs. 22–24).

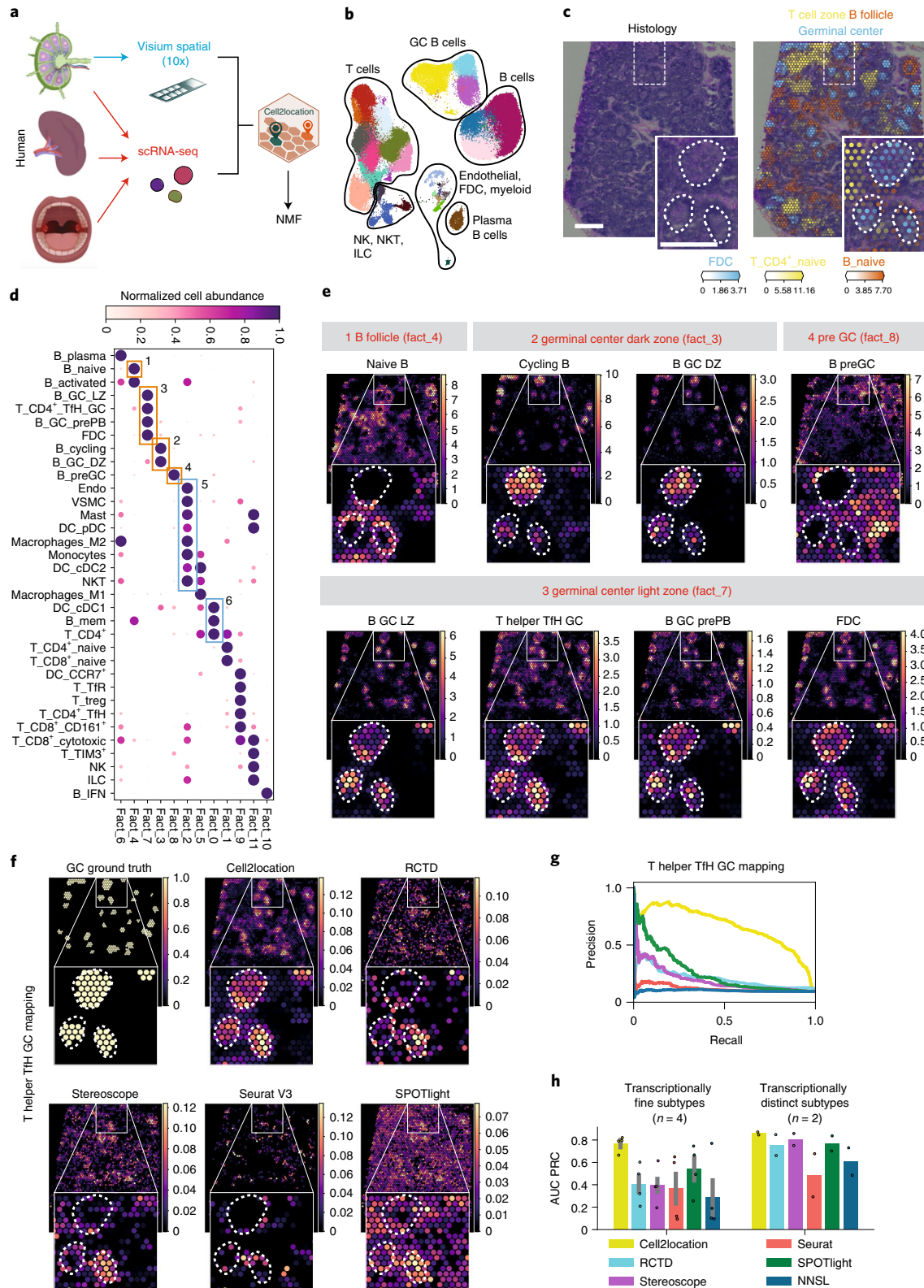
To extend our benchmarking to a second tissue with a larger number of fine-grained cell types, we examined immune cell populations in the human gut (Fig. 5a and Supplementary Figs. 25 and 26). The human gut presents a challenging application for spatial mapping as it contains a larger number of transcriptionally fine-grained immune and non-immune cell types, with the immune cells enriched in relatively small tissue areas across lymphoid follicles (for example, ~1% of tissue locations in our dataset).

We defined comprehensive cell type reference signatures by combining scRNA-seq datasets from pediatric and adult human intestinal tract, which was enriched for immune (CD45⁺) cells to

Fig. 4 | Mapping the cellular compartments of the human lymph node. **a**, Analysis workflow and downstream analysis. **b**, UMAP representation (x and y axis) of 34 immune and non-immune cell subtypes (color), combining scRNA-seq data from three studies used to define reference gene expression signatures of cell types (Methods). **c**, Major tissue zones of the lymph node, including T cell and B cell zones, and GC reactions with FDCs. Left, H&E image of the tissue section. Right, estimated cell abundance (color intensity) of cell types (color) shown over the H&E image of the lymph node sample ($n=1$ donor). Scale bar, 1 mm. **d**, Identification of lymph node cell compartments using NMF. Shown is a dot plot of the estimated NMF weights of cell types (rows) across NMF components (columns), which correspond to cellular compartments. Shown are relative weights, normalized across components for every cell type. **e**, Estimated cell abundance of selected cell types that are associated with highlighted cellular compartments (Fig. 4d) is shown in panels 1–4. Spatial plots (x and y axis) show cell abundance (color intensity) for each cell type (subpanels). Boxes highlight the tissue regions shown in inset panels. Dashed circles indicate GCs. **f**, First panel: annotations of germinal center GC locations based on paired histology image (positive locations in yellow; negative locations in black). Other panels: spatial plots (x and y axis) showing estimated cell proportion of the T_CD4⁺_Tfh_GC subpopulation (color intensity) for alternative methods (subpanels). **g**, Quantitative assessment of the mapping accuracy for the T_CD4⁺_Tfh_GC cell subpopulation using the histology annotation of GC positive and negative locations as in **f**. Shown are precision recall curves for alternative methods. **h**, Assessment of spatial mapping of six subtypes that are expected to be present in annotated GC (positive locations) and absent from other locations (as in **g**). Shown are macro-average precision recall scores (bar height, y axis) for alternative methods (color, x axis), considering either transcriptionally fine (left, $n=4$) or transcriptionally distinct (right, $n=2$) subtypes. Error bars denote standard errors. Transcriptionally fine subtypes (B_GC_DZ, B_GC_LZ, B_GC_prePB and T_CD4⁺_Tfh_GC); transcriptionally distinct subtypes (B_Cycling and FDC).

represent them at equal proportion with non-immune (CD45⁻) subtypes, spanning 76 cell types in total across 153,901 cells (Fig. 5b)⁴². We mapped these reference cell types to a Visium dataset of the adult colon consisting of two consecutive tissue sections⁴³, where we annotated the locations of lymphoid follicles using histology and marker gene expression in Visium data (Fig. 5c and

Supplementary Fig. 25b). Correspondingly, after mapping with cell2location, the expected major regions within the colon could be readily identified, including lymphoid follicles with cycling B cells, epithelial compartments with colonocytes and transit-amplifying progenitor cells as well as the muscularis mucosa layer with myofibroblasts (Fig. 5c and Supplementary Files 12 and 13)⁴³.



We identified 19 immune cell types in our gut scRNA-seq reference that are known to be enriched in lymphoid follicles^{44,45}, which we again classified into transcriptionally fine versus transcriptionally distinct cell types (Supplementary Fig. 25c) and assessed their

mapping to lymphoid follicles in the Visium data by cell2location and other methods. Although all considered methods achieved high accuracy for distinct cell types, such as cycling B cells, cell2location again achieved markedly higher accuracy for mapping fine-grained

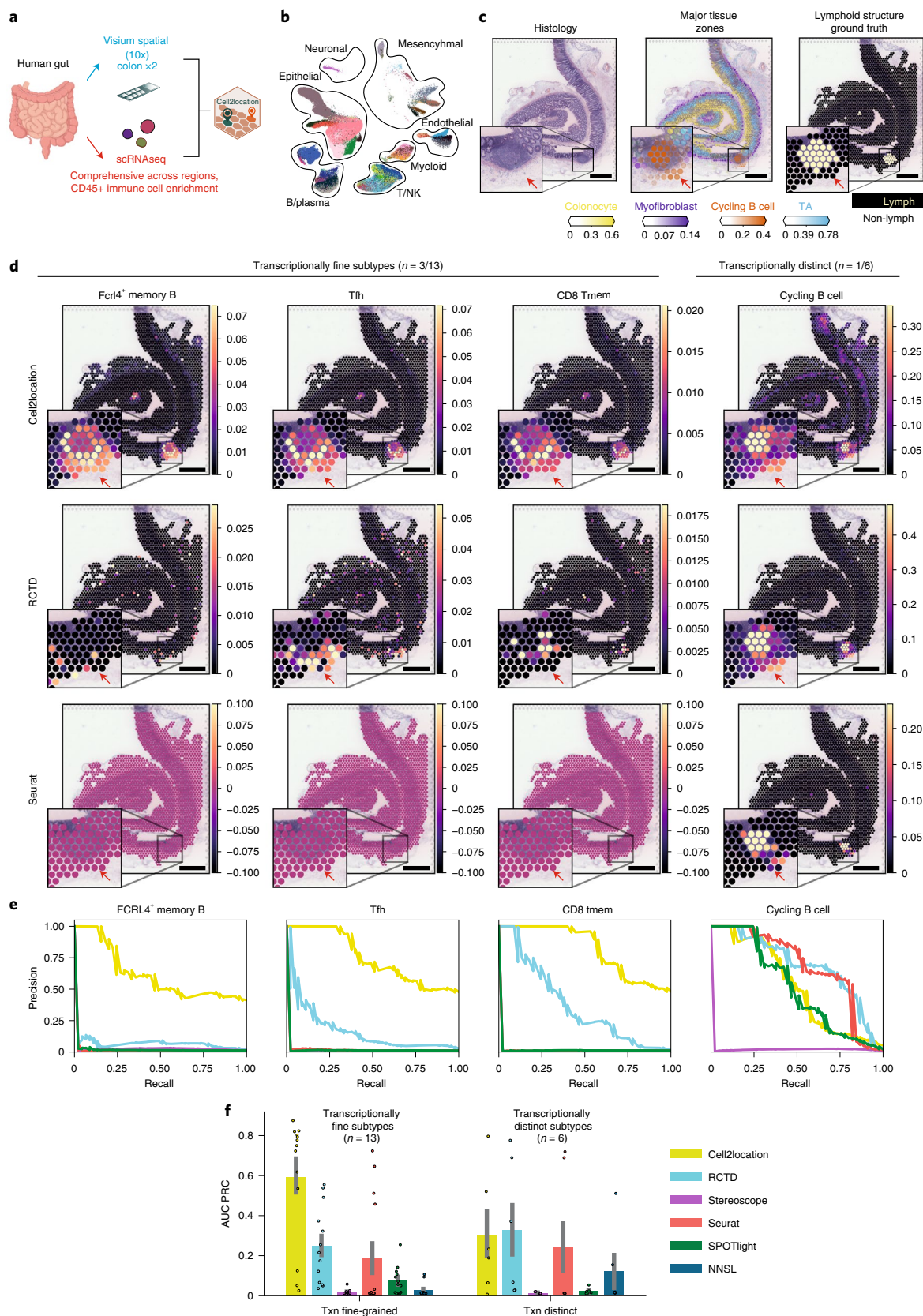


Fig. 5 | Cell2location outperforms alternative methods in mapping fine immune subtypes in the human gut. **a**, Analysis workflow. **b**, UMAP representation (x and y axis) of 76 immune and non-immune reference cell subtypes (color), which were obtained by combining scRNA-seq data from gut tissue and tertiary lymphoid follicles and used to define reference cell type signatures (Methods). **c**, Major tissue zones of the gut, including epithelial compartments with mature colonocytes and transit amplifying cells, muscularis mucosa with myofibroblasts and tertiary lymphoid follicle with immune cells (zoomed-in panel). Left, H&E image of the tissue section. Middle, estimated cell abundance (color intensity) of cell types (color) shown over the H&E image of the same gut sample. Right, ground truth histology-based annotation of lymphoid follicle (color) shown for the corresponding H&E image. $n = 1$ donor and $n = 2$ tissue sections. Scale bar, 1 mm. **d**, Spatial plots (x and y axis), depicting estimated cell type proportions (color intensity) for three selected transcriptionally fine subtypes (out of 13) and one transcriptionally distinct subtype (out of six). Cell types are shown in columns, and alternative methods to estimate the corresponding cell type proportions are shown in rows. Boxes highlight the tissue regions shown in inset panels. **e**, Precision recall curves for cell2location and alternative methods for mapping the indicated cell types for cell2location and alternative methods (methods in color; see legend in **f**). These cell types are expected to be present in annotated lymphoid follicles (positive set of locations) and absent from other locations (negative set). **f**, Summary of precision recall (AUC score, y axis) for cell2location and alternative methods (color, x axis). Cell types are stratified by the extent of transcriptional distinctiveness (fine versus distinct). These cell types are expected to be present in annotated GC (positive set of locations) and absent from other locations (negative set) for alternative methods. Shown are average AUC scores across cell types (bar height) with error bars corresponding to standard errors. AUC, area under the curve.

cell types (Fig. 5d–f; $n = 13$ cell types). Alternative methods either failed to locate several of the fine immune cell types, such as FCRL4⁺ memory B cells and Tfh cells in Visium data, or mapped them with low specificity (Fig. 5d–f and Supplementary Figs. 25d and 26).

Finally, we sought to explore the specific modeling choices in cell2location that are most relevant for the model performance (Supplementary Figs. 24 and 25e), which in particular confirmed the benefits of hierarchical modeling to borrow statistical strength across locations for mapping fine cell types.

Taken together, these results show that cell2location outperforms other methods in mapping transcriptionally fine-grained cell types in complex tissues.

Discussion

The integration of disassociated single-cell and spatial transcriptomics promises a high-throughput approach to map tissue cell types and cell signaling in situ. To best use the potential of these technologies and to create comprehensive cellular wiring diagrams of tissues, computational models for high-resolution mapping of cell types in spatial transcriptomic data are needed. Here we presented cell2location, a Bayesian model that provides sufficient sensitivity and resolution to spatially map a wide range of cell types, including transcriptionally fine subtypes, thereby allowing the creation of comprehensive atlases of diverse tissues in a versatile manner.

We performed extensive quantitative benchmarks across three tissues, consistently showing that cell2location enables a substantial advance for spatially mapping fine-grained cell types over existing tools. In the human lymph node and gut, cell2location provided higher accuracy to resolve fine immune cell populations. In the mouse brain, cell2location enabled accurate mapping of fine astrocyte subtypes.

Despite showing a small number of transcriptional differences, fine cell types often exhibit specialized and critical functions across tissues. In the immune system, we showed that cell2location can accurately resolve fine subtypes such as Tfh cells guiding B cell differentiation and FCRL4⁺ B cells modulating immune responses at mucosal sites⁴⁴. In the mouse brain, we resolved four regional astrocyte populations. In addition to their molecular and spatial identities, we can only speculate about the regionalized functions of these astrocytes. Astrocyte–neuron signaling in the lateral habenula regulates neuronal circuits implicated in depression⁴⁶. Similarly, diencephalic astrocytes could shape the activity of local neuronal circuits, such as the lateral thalamic circuitry for sensory relay or medial thalamic circuits involved in attention control⁴⁷.

Although cell2location is conceptually related to existing methods, specific design choices enable the benefits highlighted by our benchmarking results. Among the most important features, cell2location allows the borrowing of statistical strength across

locations with similar cell composition, thereby improving sensitivity for mapping fine cell types. Additionally, the model can jointly process data across batches while accounting for technical differences as well as variation in mRNA detection sensitivity. Finally, we note that cell2location is implemented as robust, fast and versatile software, integrated with scvi-tools framework¹⁹, enabling a wide range of different applications.

Although we showed that cell2location can deal with data from a range of existing technologies, upcoming technologies will motivate further refinements of the model. This includes support for higher-resolution spatial transcriptomics assays, for example by adapting the model to account for differences in the noise characteristics such as background probe binding or variable capture areas (for example, Nanostring^{48,49} and LCM-RNA-seq⁵⁰). The modular implementation of cell2location in pyro and pymc3 (refs. ^{51,52}) will facilitate such efforts.

A major future frontier for cell atlas projects is the three-dimensional (3D) mapping of tissues at cellular resolution. The spatial mapping approach described here can lay the foundations for 3D tissue atlases. Single-cell and spatial RNA-seq and imaging can be applied to consecutive tissue sections across an organ, integrating data from all technologies to build a multi-scale atlas. Regional cell type maps generated with cell2location can also guide gene selection for intact tissue volume imaging⁵³, leading to 3D models of cells in their native tissue context with whole transcriptome information.

In summary, we introduced here a versatile and flexible approach for spatial mapping of tissues. We expect that cell2location will have diverse applications in mapping tissue architecture across development, health and disease. In separate applications to the human endometrium⁵⁴, developing human gut⁴² and human gonads⁵⁵, we further showed the versatility of cell2location to map fine-grained cell type differences across diverse tissues. Going forward, spatial cell type maps coupled with cell interaction analysis can help decipher developmental processes such as cellular differentiation and how it is regulated by extrinsic cues. These developmental cues can be recapitulated in vitro to refine tissue engineering approaches. Similarly, understanding how tissue architecture is perturbed in response to disease can enrich understanding of pathological processes and aid the design of therapeutic interventions.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41587-021-01139-4>.

Received: 15 November 2020; Accepted: 28 October 2021;
Published online: 13 January 2022

References

- Stuart, T. et al. Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902 (2019).
- Rodriques, S. G. et al. Slide-seq: a scalable technology for measuring genome-wide expression at high spatial resolution. *Science* **363**, 1463–1467 (2019).
- Andersson, A. et al. Single-cell and spatial transcriptomics enables probabilistic inference of cell type topography. *Commun. Biol.* **3**, 565 (2020).
- Moncada, R. et al. Integrating microarray-based spatial transcriptomics and single-cell RNA-seq reveals tissue architecture in pancreatic ductal adenocarcinomas. *Nat. Biotechnol.* **38**, 333–342 (2020).
- Cable, D. M. et al. Robust decomposition of cell type mixtures in spatial transcriptomics. *Nat. Biotechnol.* <https://doi.org/10.1038/s41587-021-00830-w> (2021).
- Elosua-Bayes, M., Nieto, P., Mereu, E., Gut, I. & Heyn, H. SPOTlight: seeded NMF regression to deconvolute spatial transcriptomics spots with single-cell transcriptomes. *Nucleic Acids Res.* **49**, e50 (2021).
- Travaglini, K. J. et al. A molecular cell atlas of the human lung from single-cell RNA sequencing. *Nature* **587**, 619–625 (2020).
- Conde, C. D. et al. Cross-tissue immune cell analysis reveals tissue-specific adaptations and clonal architecture in humans. Preprint at https://www.researchgate.net/publication/351185031_Cross-tissue_immune_cell_analysis_reveals_tissue-specific_adaptations_and_clonal_architecture_across_the_human_body (2021).
- Vento-Tormo, R. et al. Single-cell reconstruction of the early maternal–fetal interface in humans. *Nature* **563**, 347–353 (2018).
- Zeisel, A. et al. Molecular architecture of the mouse nervous system. *Cell* **174**, 999–1014 (2018).
- Yao, Z. et al. A taxonomy of transcriptomic cell types across the isocortex and hippocampal formation. *Cell* **184**, 3222–3241 (2021).
- Lein, E., Borm, L. E. & Linnarsson, S. The promise of spatial transcriptomics for neuroscience in the era of molecular cell typing. *Science* **358**, 64–69 (2017).
- Stahl, P. L. et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science* **353**, 78–82 (2016).
- Vickovic, S. et al. High-definition spatial transcriptomics for in situ tissue profiling. *Nat. Methods* **16**, 987–990 (2019).
- Junker, J. P. et al. Genome-wide RNA tomography in the zebrafish embryo. *Cell* **159**, 662–675 (2014).
- Zeisel, A. et al. Brain structure. Cell types in the mouse cortex and hippocampus revealed by single-cell RNA-seq. *Science* **347**, 1138–1142 (2015).
- Halpern, K. B. et al. Single-cell spatial reconstruction reveals global division of labour in the mammalian liver. *Nature* **542**, 352–356 (2017).
- Aliee, H. & Theis, F. AutoGeneS: automatic gene selection using multi-objective optimization for RNA-seq deconvolution. *Cell Syst.* **12**, 706–715 (2021).
- Gayoso, A. et al. scvi-tools: a library for deep probabilistic analysis of single-cell omics data. Preprint at https://www.researchgate.net/publication/351290216_scvi-tools_a_library_for_deep_probabilistic_analysis_of_single-cell_omics_data (2021).
- Lawson, C. L. & Hanson, R. J. *Solving Least Squares Problems* (SIAM, 1995).
- Tasic, B. et al. Adult mouse cortical cell taxonomy revealed by single cell transcriptomics. *Nat. Neurosci.* **19**, 335–346 (2016).
- Phillips, J. W. et al. A repeated molecular architecture across thalamic pathways. *Nat. Neurosci.* **22**, 1925–1935 (2019).
- Lim, L., Mi, D., Llorca, A. & Marin, O. Development and functional diversification of cortical interneurons. *Neuron* **100**, 294–313 (2018).
- Tasic, B. et al. Shared and distinct transcriptomic cell types across neocortical areas. *Nature* **563**, 72–78 (2018).
- Stickels, R. R. et al. Highly sensitive spatial transcriptomics at near-cellular resolution with Slide-seqV2. *Nat. Biotechnol.* **39**, 313–319 (2021).
- Ben Haim, L. & Rowitch, D. H. Functional diversity of astrocytes in neural circuit regulation. *Nat. Rev. Neurosci.* **18**, 31–41 (2017).
- Bayraktar, O. A. et al. Astrocyte layers in the mammalian cerebral cortex revealed by a single-cell in situ transcriptomic map. *Nat. Neurosci.* **23**, 500–509 (2020).
- Batiuk, M. Y. et al. Identification of region-specific astrocyte subtypes at single cell resolution. *Nat. Commun.* **11**, 1220 (2020).
- Lanjakornsiripan, D. et al. Layer-specific morphological and molecular differences in neocortical astrocytes and their dependence on neuronal layers. *Nat. Commun.* **9**, 1623 (2018).
- Chai, H. et al. Neural circuit-specialized astrocytes: transcriptomic, proteomic, morphological, and functional evidence. *Neuron* **95**, 531–549 (2017).
- Huang, A. Y.-S. et al. Region-specific transcriptional control of astrocyte function oversees local circuit activities. *Neuron* **106**, 992–1008 (2020).
- John Lin, C.-C. et al. Identification of diverse astrocyte populations and their malignant analogs. *Nat. Neurosci.* **20**, 396–405 (2017).
- Polanski, K. et al. BBKNN: fast batch alignment of single cell transcriptomes. *Bioinformatics* **36**, 964–965 (2020).
- V1_Human_Lymph_Node - Datasets - Spatial Gene Expression - Official 10x Genomics Support. https://support.10xgenomics.com/spatial-gene-expression/datasets/1.1.0/V1_Human_Lymph_Node
- James, K. R. et al. Distinct microbial and immune niches of the human colon. *Nat. Immunol.* **21**, 343–353 (2020).
- Park, J.-E. et al. A cell atlas of human thymic development defines T cell repertoire formation. *Science* **367**, eaay3224 (2020).
- King, H. W. et al. Single-cell analysis of human B cell maturation predicts how antibody class switching shapes selection dynamics. *Sci. Immunol.* **6**, eabe6291 (2021).
- Levitin, H. M. et al. De novo gene signature identification from single-cell RNA-seq with hierarchical Poisson factorization. *Mol. Syst. Biol.* **15**, e8557 (2018).
- Kotliar, D. et al. Identifying gene expression programs of cell-type identity and cellular activity with single-cell RNA-seq. *eLife* **8**, e43803 (2019).
- Roco, J. A. et al. Class-switch recombination occurs infrequently in germinal centers. *Immunity* **51**, 337–350 (2019).
- Sun, B. T. *Helper Cell Differentiation and Their Function* (Springer, 2014).
- Elmentaite, R. et al. Cells of the human intestinal tract mapped across space and time. *Nature* **597**, 250–255 (2021).
- Fawcner-Corbett, D. et al. Spatiotemporal analysis of human intestinal development at single-cell resolution. *Cell* **184**, 810–826 (2021).
- Spencer, J. & Sollid, L. M. The human intestinal B-cell response. *Mucosal Immunol.* **9**, 1113–1124 (2016).
- Koboziev, I., Karlsson, F. & Grisham, M. B. Gut-associated lymphoid tissue, T cell trafficking, and chronic intestinal inflammation. *Ann. N. Y. Acad. Sci.* **1207**, E86–E93 (2010).
- Cui, Y. et al. Astroglial Kir4.1 in the lateral habenula drives neuronal bursts in depression. *Nature* **554**, 323–327 (2018).
- Ward, L. M. The thalamus: gateway to the mind. *Wiley Interdiscip. Rev. Cogn. Sci.* **4**, 609–622 (2013).
- Merritt, C. R. et al. Multiplex digital spatial profiling of proteins and RNA in fixed tissue. *Nat. Biotechnol.* **38**, 586–599 (2020).
- Roberts, K. et al. Transcriptome-wide spatial RNA profiling maps the cellular architecture of the developing human neocortex. Preprint at https://www.researchgate.net/publication/350237379_Transcriptome-wide_spatial_RNA_profiling_maps_the_cellular_architecture_of_the_developing_human_neocortex (2021).
- Nichterwitz, S. et al. Laser capture microscopy coupled with Smart-seq2 for precise spatial transcriptomic profiling. *Nat. Commun.* **7**, 12139 (2016).
- Salvatier, J., Wiecki, T. V. & Fonnesbeck, C. Probabilistic programming in Python using PyMC3. *PeerJ Comput. Sci.* **2**, e55 (2016).
- Bingham, E. et al. Pyro: deep universal probabilistic programming. *J. Mach. Learn. Res.* **20**, 973–978 (2019).
- Zhao, S. et al. Cellular and molecular probing of intact human organs. *Cell* **180**, 796–812 (2020).
- Garcia-Alonso, L., Handfield, L. F. & Roberts, K. et al. Mapping the temporal and spatial dynamics of the human endometrium in vivo and in vitro. *Nat. Genet.* **53**, 1698–1711 (2021).
- Vento-Tormo, R. et al. Single-cell roadmap of human gonadal development. Preprint at https://assets.researchsquare.com/files/rs-496470/v1_covered.pdf?c=1631867880 (2021).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2022

Methods

Cell2location model. For a complete derivation of the cell2location model, see the Supplementary Methods. In brief, cell2location is a Bayesian model that estimates the absolute abundance of cell types at each location by decomposing the spatial expression count matrix into a predefined set of reference cell type signatures. The model takes an untransformed spatial expression count matrix $d_{s,g}$ of genes $g = \{1, \dots, G\}$ at locations $s = \{1, \dots, S\}$ as input, as, for example, obtained from the 10x Space Ranger software (10x Visium data). Cell2location allows the joint modeling of multiple spatial datasets $e = \{1, \dots, E\}$ (also known as batch integration) where e denotes individual experiments obtained using one or multiple technologies. The second input is a matrix of reference cell type signatures $g_{f,g}$, which correspond to the expected mRNA count of genes g for each cell type $f = \{1, \dots, F\}$. By default, this matrix is estimated using a negative binomial regression model, which allows for combining data across batches and technologies (see below and Supplementary Methods).

Cell2location model. Cell2location models the elements of the spatial expression count matrix $d_{s,g}$ as negative binomial distributed, given an unobserved gene expression level (rate) $\mu_{s,g}$ and gene- and batch-specific over-dispersion $\alpha_{e,g}$:

$$d_{s,g} \sim NB(\mu_{s,g}, \alpha_{e,g}).$$

The expression rate of genes g at location s , $\mu_{s,g}$, in the mRNA count space is modeled as a linear function of reference cell types signatures $g_{f,g}$:

$$\mu_{s,g} = \left(\underbrace{m_g}_{\text{technology sensitivity}} \cdot \underbrace{\sum_f w_{s,f} g_{f,g}}_{\text{cell type contributions}} + \underbrace{s_{e,g}}_{\text{additive shift}} \right) \cdot \underbrace{y_s}_{\text{per-location sensitivity}}.$$

Here, $w_{s,f}$ denotes regression weight of each reference cell type f at location s , which can be interpreted as the abundance of cells expressing reference cell type signature f at location s ; m_g is a gene-specific scaling parameter that adjusts for differences in sensitivity between technologies; $s_{e,g}$ captures gene-specific additive shift that allows the model to account for contaminating RNA; and y_s is a location-specific scaling parameter that models variation in RNA detection sensitivity across locations s across one or multiple batches e (by use of hierarchical priors; Supplementary Methods).

Because of the organization of tissues into zones, the cell type abundance estimates $w_{s,f}$ are not independent across locations. To borrow statistical strength and use information sharing across locations, $w_{s,f}$ is modeled using a hierarchical decomposition prior (factorization), assuming $r = \{1, \dots, R\}$ groups of cell types that partially share a common cell abundance profile (for example, due to tissue zones; Supplementary Methods). Unless stated otherwise, R is set to 50.

Approximate variational inference is used to estimate all model parameters, implemented in the scvi-tools framework¹⁹, building on pyro and pymc3 probabilistic programming frameworks as backend^{21,22}, which supports GPU acceleration. For full details, see the Supplementary Methods.

Note on the construction of reference cell type signatures. It is important to aim for a comprehensive and detailed cell-type reference, which includes as many of the cell types and subpopulations that are present in situ as possible. This can be achieved by generating a paired snRNA-seq reference from the same tissue sample. However, imperfect matching of cell populations is often acceptable (Figs. 4 and 5 and Supplementary Fig. 4c).

Based on one or multiple snRNA-seq input datasets, reference cell type signatures (gene expression) are estimated based on a defined set of cell populations, which, in turn, can be identified using conventional clustering workflows. By default, the cell2location employs a negative binomial regression to estimate reference cell type signatures. This model-based approach allows robust combination of data across technologies and batches (Supplementary Fig. 19), which results in improved spatial mapping accuracy (Supplementary Figs. 24a–c and 25e). Alternatively, if batch effects are not a concern, a hard-coded method that estimates per-cluster average mRNA counts without any model-based strategy can be sufficient. This strategy is simple and fast and yields similar levels of accuracy if batch effects are small (Supplementary Fig. 24a–c). The hard-coded method is also recommended for non-unique molecular identifier (UMI) technologies for which the negative binomial count model is often not appropriate, such as Smart-seq2.

Note on determining model hyper-parameters. The cell2location model has two parameters that should be adjusted by the user, taking known experimental and biological characteristics of a given dataset into consideration:

1. Expected cell abundance per location $\hat{N}$
2. Regularization of within-experiment variation in RNA detection sensitivity α'

Supplementary Fig. 27 provides a flowchart to guide the user to determine appropriate settings for these hyper-priors, which were used to determine the model parameters for all presented results.

The expected cell abundance $\hat{N}$ per location is provided as a single tissue-level estimate, which can be derived from histology images (H&E or DAPI), ideally paired to the spatial expression data or at least representing the same tissue type. This parameter can be estimated by manually counting nuclei in 10–20 locations in the histology image (for example, using the 10x Loupe browser; Supplementary Fig. 8) and computing the average cell abundance. An appropriate setting of this prior is essential to inform the estimation of absolute cell type abundance values; however, the model is robust to a range of similar values (Supplementary Fig. 5). In settings where suitable histology images are not available, the size of capture regions relative to the expected size of cells can be used to estimate $\hat{N}$ (Slide-seq V2; Supplementary Fig. 27). For all analyses in this manuscript, a single tissue-level estimate was used; however, as an advanced feature, cell2location can use the per-location number of cells.

Hyper-parameter α' for regularizing within-experiment variation in RNA detection sensitivity. α' is the shape parameter of the prior gamma distribution over y_s that represents the extent to which RNA detection sensitivity for individual locations deviates from the mean sensitivity of each batch. Large values of α' correspond to small within-experiment variability (Supplementary Methods). By default, we assume small within-experiment variability in RNA detection sensitivity across locations ($\alpha' = 200$), which results in values of y_s close to the mean sensitivity for each experiment. However, when strong gradients in mRNA detection sensitivity are observed, which is a common issue in adult human 10x Visium data, we recommend less strict regularization ($\alpha' = 20$).

Although good choices of these hyper-parameters can have a positive effect on accuracy and sensitivity, the estimates are robust to a range of suboptimal choices (Supplementary Fig. 5). The estimation of absolute cell abundance requires appropriate settings of $\hat{N}$ in particular.

Validation of cell2location using synthetic data and comparison to alternative methods.

Constructing a synthetic spatial transcriptomics dataset. Simulated spatial transcriptomics data were generated by combining expression profiles of cells drawn from each of 49 cell types in the mouse brain snRNA-seq reference data (see below), generating abundance profiles at 2,500 locations. snRNA data from the two most mouse brain snRNA-seq samples that had minimal batch effect were split into one dataset used to generate the synthetic data (50% of cells) and a second dataset used to evaluate cell2location and alternative methods (50% of cells), similarly to the strategy proposed by Andersson et al.¹. Hyper-parameters for data simulation were chosen to mimic (the typically low) cell counts observed for cell types in real tissues, additionally matching empirically observed sparsity profiles. Cell type abundances were simulated according to either a spatially ubiquitous pattern (eight cell types) or a regional pattern (41 cell types). Regional patterns are represented by 12 tissue zones defined by co-located cell types that mimic the organization of real tissues. The assignment of 41 regional cell types to the 12 tissue zones is shown in Supplementary Fig. 2, with each cell type belonging to 1–3 tissue zones and each tissue zone containing 2–8 cell types. The number of cell types present at each location, as well as the absolute abundance (the number of cells per location), were simulated according to either low or high average cell type abundance (Fig. 1b), stratified by ubiquitous and regional location patterns (see below). The mathematical description and the step-by-step procedure for simulating abundance of cell types across locations and generating multi-cell mRNA counts is described in detail in the Supplementary Methods (section 5).

Validation of cell2location using synthetic data. Synthetic gene expression data were constructed as described above to assess the performance of cell2location and alternative methods. The synthetic data provide ground truth values for the relative cell abundance of each cell type (cell type proportions), absolute cell abundance (including partial cells) and total mRNA count of each cell type for each location, which were used to benchmark estimates obtained from cell2location and alternative methods. All considered models were provided with the ground truth snRNA-seq reference data (evaluation subset, see above; Table 1) as well as the synthetic mRNA count matrices for all locations.

Cell2location. Signatures of reference cell types were derived using a negative binomial regression (Supplementary Methods) with default parameters, except training mini-batch size of 2,500 and training duration of 600 epochs. The hyper-parameter values used for the synthetic data analysis are shown in Table 1. In addition, the sensitivity of cell abundance estimates to the choices of these hyper-parameters was assessed (Supplementary Fig. 5, section on sensitivity and robustness below). The prior cell abundance $\hat{N} = 9$ was set to a value close to, but not fully matching, the ground truth ($\hat{N} = 12$), with the aim to obtain as realistic results as possible.

Assessment of alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS implementation) were used with default or optimized parameter settings as described below. Stereoscope training hyper-parameters were adapted to each dataset as stated in the respective sections. All benchmark datasets were provided with the identical cell type reference snRNA-seq data and spatial count matrix as used for cell2location. The same ground truth relative cell abundance is used for all cell types.

Table 1 | Datasets used in the paper and corresponding cell2location hyper-parameters

	Spatial data	<i>n</i> samples	<i>n</i> locations	Reference data	<i>n</i> cell types	<i>n</i> cells	<i>n</i> genes	Cell abundance prior ($\bar{N}$)	Normalization (γ_i) alpha prior	<i>n</i> iterations
1	Synthetic	1	2,500	Paired (our) mouse brain snRNA-seq	49	8,111	12,080	9	200	40,000
2	Mouse brain Visium	5	14,968	Paired (our) mouse brain snRNA-seq	59	40,532	12,809	8	200	20,000
3	Mouse brain Visium (SSp cortex locations)	5	1,393	Yao et al., SSp cortex and hippocampus	121	6,948	13,565	8	200	40,000
4	Slide-seq V2 (section 2, hippocampus)	1	33,108	Our mouse brain snRNA-seq	59	40,532	10,670	1	20	30,000
6	Human lymph node (10x)	1	4,035	Combined lymph tissue reference	34	73,260	10,217	30	200	30,000
7	Adult gut (Fawcner-Corbett et al.)	2	3,842	Adult and pediatric scRNA-seq (Elmentaite et al.)	76	153,901	14,388	12	20	20,000

All other parameters were used at their default values (Supplementary Methods). *n* genes denote the number of genes shared between scRNA and spatial data.

Stereoscope³. This method uses a model-based approach to estimating both the cell type signatures and the relative cell abundance across locations (see the comparison in the Supplementary Methods). A negative binomial observation model is parameterized using a gene- and cell-type-specific log-odds parameter and a gene-specific total count parameter. This custom signature estimation method is hard-coded into the workflow, and both estimated parameters are used as input to the model for estimating relative cell abundance. Unless stated otherwise, default values of hyper-parameters are used. The scvi-tools¹⁹ implementation of the Stereoscope model was used with the following hyper-parameters (default scvi-tools values were modified to achieve improved accuracy):

- No highly variable gene (HVG) selection was used (12,054 genes shared between single-cell and spatial data)
- Estimation of expression signatures of cell types: 0.01 learning rate; 1,500 epochs (increased compared to default to observe convergence); 1,024 mini-batch training size (similar to original Stereoscope default values).
- Estimation of relative cell abundance: 0.01 learning rate; 30,000 epochs (also increased compared to default); 1,250 mini-batch training size (half of the total number of locations, similar to original Stereoscope default values).

RCTD⁵. This method uses a similar model-based approach to the estimation of relative cell abundance across locations (see the comparison in the Supplementary Methods). It estimates reference cell type signatures by computing average mRNA count per cluster (hard-coded). RCTD was used in default mode (not 'doublet' mode) with hyper-parameters selected as suggested in the GitHub tutorial, with a reduced threshold of the number of cells per cell type to ensure that all 49 cell types were considered and including gene filtering based on log fold change (default settings of all other parameters; <https://github.com/dmccable/RCTD/tree/7a895ed1666269784cc472fa0a6e4abbd3d2c2b2>).

Seurat V3 (principal component analysis, anchor method and mutual nearest neighbor classifier). Seurat V3 workflow was used to project synthetic locations into the principal component analysis (PCA) space constructed using snRNA-seq reference data (Seurat V3 with anchor selection, Seurat::FindTransferAnchors), with subsequent transfer of cell type labels onto synthetic locations using a mutual nearest neighbor (MNN) classifier (Seurat::TransferData). For the purpose of this comparison, the label transfer scores were interpreted as relative cell type abundance estimates of each cell type for each location. Before integration, both snRNA-seq reference and synthetic spatial datasets were subjected to SCTransform normalization (Seurat::SCTransform)⁵⁶. The PCA method (with default 30 components) was used for integration (Seurat::FindTransferAnchors) as advised in Seurat package documentation and spatial mapping tutorial. All functions were used with default parameters.

SPOTlight⁶. SPOTlight uses a seeded topic model to find expression signatures of reference cell types and a modified non-negative least squares (NNLS) method (denoted NMFreg) to estimate relative cell abundance. The topic model is initialized in a way that guides the topics to resemble expression signatures of cell types. The default SPOTlight workflow was implemented with recommended parameters

(<https://github.com/MarcElosua/SPOTlight/tree/fb5e7c7de3e6d0ac7618c4b1a55aea06e5f472ea>).

In brief, to improve computational efficiency, the snRNA-seq data size was reduced by randomly selecting, at most, 100 cells per cell type (when a cell type

is represented by fewer than 100 cells, all cells are kept), and highly variable genes (5,000 genes) were selected using Seurat::FindAllMarkers tool. Untransformed and unnormalized expression counts for both snRNA-seq reference and synthetic locations were provided as input to the 'spotlight_deconvolution' function in the SPOTlight package with the following hyper-parameters: transf = 'uv', method = 'nsNMF', and, crucially, 'minimum allowed proportions of cell types' min_cont = 0.01.

NNLS (AutoGeneS)^{18,20}. NNLS was used to estimate the absolute cell abundance of each cell type in each synthetic location given the reference signatures of cell types, which were derived by computing average expression of each gene across cells of each type. Untransformed and unnormalized expression counts of the synthetic locations were used as input to NNLS implementation as part of the AutoGeneS package. Similarly to cell2location, the output was normalized to produce relative cell abundance.

Comparison of cell2location and alternative methods. The estimates of relative cell abundance from all methods were compared to the ground truth using two-dimensional (2D) histogram plots, Pearson *R* (numpy.corrcoef) and Jensen-Shannon divergence (scipy.spatial.distance.jensenshannon) and reported in Fig. 1b and Supplementary Fig. 3a. In addition, the accuracy was assessed using precision recall quantified on all 49 cell types jointly (macro-average). Relative cell abundance was used as a predictor, and cell abundance >0.1 was used as a gold standard label. Precision recall results are reported in Supplementary Fig. 3b. To simplify presentation, results across individual cell types were summarized as an average Pearson *R* (Fig. 1c and Supplementary Fig. 4b) or precision recall score (S3B and S4B) and shown as bar plots, with bar height denoting average values across cell types and error bars denoting standard errors of the mean. To assess absolute cell abundance estimation, estimated cell abundance was compared to simulated cell abundance using 2D histogram plots, Pearson *R* (numpy.corrcoef) and Jensen-Shannon divergence (scipy.spatial.distance.jensenshannon) and reported in Supplementary Fig. 4a.

Sensitivity of cell2location modeling choices and robustness. *Assessment of the effect of modeling choices.* The effect of cell2location modeling choices on the performance of the model was evaluated using both synthetic (Supplementary Fig. 4b) human lymph node data (Supplementary Fig. 24) and human gut data (Supplementary Fig. 25e). In brief, the relevance of the following modeling choices (for details, see Supplementary Methods, section 3) was evaluated using the benchmarking criteria as defined in the main analysis of the respective datasets (Pearson *R* of cell type proportions and precision recall curves of cell type presence/absence):

1. Prior factorization of w_{ij}
2. Gene-specific technology sensitivity scaling m_{ij}
3. Negative binomial regression for estimating reference signatures of cell types versus a hard-coded method

Assessment of the robustness of the cell abundance estimates when removing increasing fractions of cell types from the reference data. Cell2location was assessed using synthetic data combined from 49 cell types (Supplementary Fig. 4c, see above). The accuracy was assessed using precision recall (macro-average) for 17 cell types that were retained in the reference when considering alternative removal schemes to subsample the full set of cell type references. Relative cell abundance was used as a predictor, and cell abundance >0.1 was used as a gold standard label.

Assessment of the robustness to hyper-parameter settings. To assess the sensitivity of cell2location with respect to selected parameters (Supplementary Fig. 5 and Supplementary Methods), the benchmark analysis on both synthetic and gut data was performed as described in the corresponding sections. Evaluated were the model performance when varying the two user-defined tissue hyper-parameters (expected cell abundance per location $\hat{N}$ and regularization of within-experiment variation in RNA detection sensitivity α') as well as hyper-parameters that govern the hierarchical prior of the decomposition across locations (Supplementary Methods). Hyper-parameters were varied one at a time while retaining the values of the three remaining hyper-parameters constant. Additionally, we assessed the effect of α' hyper-parameter on the consistency of estimated total cell abundance by cell2location and the nuclei counts estimated by segmenting histology images in the mouse brain (Supplementary Fig. 8f and Supplementary Methods), as quantified by Pearson correlation and root mean squared error.

Mouse brain experiments and analysis. *Mouse brain tissue processing.* Tissue processing. Mouse work was carried out at the Wellcome Sanger Institute in accordance with UK Home Office regulations and the UK Animals (Scientific Procedures) Act of 1986 under a UK Home Office license, which were regularly reviewed by the institutional Animal Welfare and Ethical Review Body.

Brains of wild-type adult C57BL/6 mice (postnatal day 56, one female and one male) were dissected, snap-frozen, embedded in optimal cutting temperature (OCT) compound (Tissue-Tek) and stored at -80°C . Brain hemispheres were cryosectioned at -20°C using a cryostat (Leica, CM3050S). To assess tissue quality, RNA was extracted from test tissue sections using the RNeasy Pico Kit (Qiagen) and yielded high RNA integrity number values (9.6 and 9.7) on an Agilent Bioanalyzer, indicating high RNA quality.

For matched single-nuclei and Visium RNA-seq experiments, brain hemispheres were cryosectioned to adjacent thick (200- μm) and thin (10- μm) coronal sections, respectively, and processed the same day. In total, four consecutive sets of thick and thin tissue sections were collected from each brain. Five sets of tissue sections yielded good-quality single-nuclei data as well as good-quality Visium data (three adjacent sections from mouse 1 and two sections from mouse 2), and one additional section from mouse 2 yielded good-quality single-nuclei data; these were considered for analysis in this study.

snRNA-seq. Thick (200- μm) mouse brain sections were cryosectioned, dissected from OCT and kept in a tube on dry ice until subsequent processing. Nuclei were extracted from each section as described previously³⁷. In brief, nuclei were released from sections via Dounce homogenization, Hoechst-stained and isolated via fluorescence-activated cell sorting. Nuclei were then loaded into the 10x Chromium Single Cell 3' Kit (v3) to obtain 3,000–7,000 nuclei per well, and library preparation was done per the manufacturer's protocol. Libraries were sequenced on an Illumina NovaSeq S4 system.

Visium spatial transcriptomics. Thin (10- μm) mouse brain sections were cryosectioned and mounted directly onto separate capture areas on 10x Visium Spatial Gene Expression slides (beta product version). Processing was done per the manufacturer's protocols. In brief, sections were methanol-fixed, H&E-stained and imaged on a NanoZoomer 2.0 slide scanner (Hamamatsu). Sections were then permeabilized and further processed to obtain cDNA libraries that were quality controlled using the Agilent Bioanalyzer. The cDNA libraries were sequenced on the Illumina HiSeq 4000 system, aiming at 300 million raw reads per section with read lengths 28cy R1, 8cy i7 index, 0cy i5 index and 91cy read 2.

Single-nucleus data processing and cell type annotation. Sequencing data were processed using 10x Cell Ranger version 3.0.2, aligned to mouse pre-mRNA genome reference version mm10 (downloaded from the 10x website), and mRNA count matrices were generated by adding intronic and exonic UMI counts for each gene in each cell. Initially, snRNA-seq counts were processed using standard Seurat V3 workflow without correcting batch effects among six individual samples. Specifically, the expression matrices filtered to include droplets containing cells (10x Cell Ranger output), were imported into the Seurat object. Cells were filtered using the following quality control criteria: number of detected genes per cell $<5,000$, fraction of nuclear-encoded mitochondrial genes <0.10 and doublet scores (scrublet³⁸) <0.2 . The data were then subjected to normalization by a scale factor of 10,000 and log1p-transformed (Seurat::NormalizeData), scaled gene-wise by subtracting the mean and divided by the standard deviation (Seurat::ScaleData). After that, 50 principal components were found (Seurat::RunPCA) and used as input to the uniform manifold approximation and projection (UMAP) algorithm (Seurat::RunUMAP), which was also used for k -nearest neighbor (KNN) graph construction with $k=100$ (Seurat::FindNeighbors). To generate a comprehensive cell annotation, joint Louvain clustering with resolution 7 was performed (Seurat::FindClusters) on all samples, which yielded 51 cell clusters. Clusters were subsequently annotated using marker genes from the literature and existing mouse brain single-cell reference datasets^{10,16,21}. To annotate the regional identity of cell types, we used (1) known regional marker genes^{10,16,21}, (2) the spatial expression patterns of these genes on in situ hybridization data from the Allen Brain Atlas (<https://mouse.brain-map.org/>) and (3) the regional localization of cell types

estimated by cell2location (Fig. 2 and Supplementary Files 1–5). The marker genes of regional cell types reported in Supplementary Fig. 7 were obtained from Zeisel et al.¹⁰ (<http://mousebrain.org/downloads.html>, L5_Allagg.loom file). This step yielded an annotation for 44 of 51 clusters, including known regional neuronal and glial subtypes as described above. The remaining seven clusters were composed of low-quality cells or unknown cell types and not analyzed further. Next, to resolve fine-grained astrocyte subtypes, a second targeted analysis was performed using cells from the two astrocyte clusters (2/44) as described below. Collectively, these two steps resulted in reference gene expression signatures for 59 distinct cell subtypes. This integrated reference was used throughout the manuscript to avoid presenting spatial mapping results with varying granularity of astrocyte subtypes. The resulting clusters and cell annotations are reported in Fig. 2b and Supplementary Figs. 6 and 7 and were used as input to cell2location and alternative methods.

Identification of astrocyte subtypes using initial spatial mapping in the mouse brain. To identify regionally enriched astrocytes, we annotated and spatially mapped reference signatures of astrocyte subpopulations. Cell clusters (2/44) annotated as astrocytes were selected from the initial analysis (Fig. 2b), as identified based on the expression of canonical markers (for example, *Aldoc* and *Slc1a3*). These cells were subjected to analysis workflow in scanpy³⁹ as follows. First, an untransformed and unnormalized count matrix was filtered to select expressed genes at two cutoffs:

1. Selecting genes detected at mRNA count >0 in $>5\%$ of cells
2. Selecting genes detected at mRNA count >0 in a few cells ($5\% > \text{cell count} > 0.4\%$) but with a mean expression >1.122018 across cells with count >0

The filtered data were then log1p-transformed and scaled. PCA was performed (30 components), and the first component, which was strongly associated with total mRNA count per cell, was removed. Next, BBKNN³³ was used to refine the integration across all six snRNA-seq datasets with the default parameters, except neighbors_within_batch = 3. The corrected KNN graph was used to perform UMAP dimensionality reduction and Leiden clustering with resolution 2.4, identifying 17 astrocyte subclusters (Supplementary Fig. 13). Marker genes of these subtypes were identified using differential expression tools in the scanpy package and markers known from the literature.

Next, an initial mapping with cell2location was performed using hyper-parameters described below in the mouse brain Visium section. Reference cell type signatures for initial mapping included 17 astrocyte subclusters as well as all other non-astrocyte cell types shown in Fig. 2b. Clusters that were not sufficiently distinct in their location (Supplementary Fig. 14) and marker genes (Supplementary Fig. 13) were merged. This resulted in ten molecularly and spatially distinct astrocyte subtypes (Fig. 3 and Supplementary Fig. 15b,c). These subtypes were incorporated into cell annotation used throughout the paper.

Visium data processing. 10x Visium spatial sequencing data were aligned to mouse pre-mRNA genome reference version mm10 (downloaded from the 10x website) using 10x Space Ranger 1.0, and mRNA count matrices were generated by adding intronic and exonic reads for each gene in each location. The paired histology H&E images were processed using 10x Space Ranger to select locations covered by tissue by aligning pre-recorded spot locations with fiducial border spots in the histology image. This allows evaluating the correspondence between cell maps produced using our method and the known brain anatomy. This also allows identifying the number of nuclei in each spot using nuclear segmentation, as described in the Supplementary Methods and reported in Supplementary Fig. 8a–d. The histology image was used to manually annotate cortical layers in the primary SSp cortex region using the lasso tool in the 10x Loupe browser.

Spatial mapping of mouse brain cell types to 10x Visium data. Spatial mapping using cell2location. Expression signatures of 59 mouse brain reference cell types (Fig. 2b, see above, and Table 1) were derived using a negative binomial regression model (Supplementary Methods) with training mini-batch size of 2,500 and training duration of 250 epochs. Mouse brain 10x Visium data were processed as described above to untransformed and unnormalized mRNA counts and filtered to genes shared with scRNA-seq (see Table 1 for gene and location count), which were used as input to cell2location with hyper-parameters as stated in Table 1.

Cell2location estimates described in this section are reported in Figs. 2d–g and 3b–e, Supplementary Figs. 8c–e, 9, 10, 12b,c, 14, 15 and 18 and Supplementary Files 1–6.

Region identification. Estimated absolute cell abundance was used to calculate a KNN graph (n neighbors = 38) followed by Leiden clustering of locations (resolution = 1.3), yielding 31 clusters. The clusters were annotated by comparing the paired histology images, and clusters marking anatomically similar regions were merged, yielding 29 distinct clusters annotated as tissue regions and reported in Fig. 3b and Supplementary Figs. 10 and 15c.

Assessment of cell2location and alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS) were used with default parameters as described in

the section on synthetic data (Methods). Stereoscope (implementation using scvi-tools; see Methods, synthetic data section) was used with the following hyper-parameters:

1. No HVG selection was used (12,809 genes shared between single-cell and spatial data)
2. Estimating expression signatures of cell types: learning rate of 0.01 (default); 1,000 epochs (increased compared to default to achieve convergence); batch size of 256 (default)
3. Estimating relative cell abundance in spatial data: learning rate of 0.01 (default); 20,000 epochs (increased compared to default to achieve convergence); batch size of 256 (default)

Assessment of cell2location and alternative methods on the mouse brain was qualitative and is presented in Supplementary Fig. 18.

Analysis of Slide-seq V2 data. Expression signatures of 59 mouse brain reference cell types (Fig. 2b) were derived using the negative binomial regression model as described above. Slide-seq V2 data of the mouse brain originating from the region containing the hippocampus, partial cortex and thalamus were downloaded from the data portal for Stickels et al.²⁵. Genes with mRNA count >0 in at least 500 locations and locations with at least 200 UMI counts were retained to reduce data size. One Slide-seq V2 section with the largest number of spatial locations was used as input to cell2location using hyper-parameters stated in Table 1. Cell2location estimates described in this section are reported in Fig. 2h, Supplementary Fig. 12 and Supplementary Files 8 and 9.

smFISH. A wild-type adult C57BL/6 mouse (postnatal day 63, male) was transcardially perfused with ice-cold PBS and 4% paraformaldehyde (PFA) in 1× PBS. The brain was dissected and post-fixed in 4% PFA at 4°C overnight and cryoprotected in 30% sucrose for 48 h at 4°C. The brain was then embedded in OCT and frozen on isopentane dry ice slurry and stored at −80°C. Coronal cryosections (12-μm) were collected on a cryostat (Leica), mounted onto superfrost glass slides (VWR) and stored at −80°C.

RNA-seq smFISH was performed as previously described⁶⁰. Tissue slides were thawed at room temperature for 15 min and baked at 60°C for 30 min. Afterwards, slides were post-fixed for 15 min in chilled 4% PFA and serially dehydrated through 50%, 70%, 100% and 100% ethanol for 5 min each. Four-plex RNA-seq smFISH was automated using a Leica BOND RX (LS Multiplex Assay, Advanced Cell Diagnostics, Bio-Techne). Tissue sections were permeabilized using heat (5 min in Leica BOND ER2 buffer at 95°C) and protease digestion (Advanced Cell Diagnostics Protease III for 20 min). The probes in C1, C2 and C3 channels were labeled using Opal 520, 570 and 650 fluorophores (PerkinElmer, diluted 1:1,500), respectively. The C4 probe was developed using TSA biotin (TSA Plus Biotin Kit, PerkinElmer, 1:500) and streptavidin-conjugated Atto 425 (Sigma-Aldrich, 1:400). Nuclei staining was performed with DAPI (Life Technologies, 1:50,000).

Imaging was performed on an Opera Phenix High-Content Screening System (PerkinElmer) in spinning disk confocal mode using a ×20 water immersion objective (NA 0.16, 298.99 nm per pixel) with a 1-μm z-step size. The excitation (Ex) laser and emission (Em) filter wavelengths were as follows: DAPI (375 nm; 435–480 nm), Atto 425 (425 nm; 463–501 nm), Opal 520 (488 nm; 500–550 nm), Opal 570 (561 nm; 570–630 nm) and Opal 650 (640 nm; 650–760 nm). Illumination correction and stitching were performed on maximum z-projection images using Acapella scripts provided by PerkinElmer.

Astrocyte segmentation and smFISH quantification. The segmentation of astrocytes was performed on the Agt image channel with pixel classification of ilastik⁶¹. The Agt channel was Gaussian blurred with a sigma of 2 pixels, and the astrocyte mask was subsequently generated by the trained pixel classifier. The performance of this classifier was evaluated by using visual inspection and a metric called out-of-bag (oob) error, which measures prediction error by using bootstrap aggregating. The oob error of the classifier was 2.857%, measured on a training image of dimension 402 × 2,010 pixels. As Agt is expressed at low levels in astrocytes outside the diencephalon²⁷, we manually traced the diencephalon to filter all astrocyte signals outside.

Two rounds of ‘closing’ operations were then executed to connect as many major astrocyte cellular processes to the cell bodies as possible. As a consequence, some cells were erroneously fused together. To split these fused cells, we performed six rounds of opening on large cells (>10,000 pixel²). Finally, all small partially segmented cells (<250 pixel²) were filtered out, and the mean pixel intensities (sum intensity/area) of target gene channels were quantified per cell.

Assessment of alternative methods at mouse brain cell type mapping with region-specific scRNA-seq reference. Construction of single-cell reference data. For quantitatively validating the cell type maps produced by cell2location, we considered a publicly available¹¹ gold standard reference of cortical and hippocampal cell locations. In that data, individual layers of the cortex and the hippocampus are dissected and subjected to scRNA-seq using the SMART-Seq 4 protocol (Fig. 2i). Data from this study were obtained from <https://brain-map-portal-stage.us.aldryn.io/atlas-and-data/rnaseq>. Cells originating from the primary SSp cortex (number of positive cell types, $n = 98$) as well as

exclusive to the hippocampus (the subiculum areas indicated as par-post-pre, number of negative cell types, $n = 23$) were selected, discarding cell types that contribute fewer than three cells. The count of cells in this scRNA-seq data was used to compute the proportion of each reference cell type in each cortical layer (Supplementary Fig. 11b), which can serve as a complementary approach to estimate relative cell type abundance (see below). The corresponding reference signatures of 121 cell types were derived as per-cluster average expression (hard-coded method), as the negative binomial regression model is not suitable for SMART-Seq data (Supplementary Methods).

Spatial mapping using cell2location. We mapped cell types from this reference to a subset of the mouse brain 10x Visium data, restricted to the primary SSp cortex (which was annotated based on histology; Supplementary Fig. 11). Data were processed as described in the section on the mouse brain, and mRNA counts were used as input to cell2location using the hyper-parameters stated in Table 1.

Assessment of cell2location and alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS) were used with default parameters as described in the section on synthetic data (Methods). Stereoscope (implementation using scvi-tools; see Methods, synthetic data section) was used with the following hyper-parameters:

4. No HVG selection was used (13,565 genes shared between single-cell and spatial data).
5. Estimating expression signatures of cell types: learning rate of 0.01 (default), 2,000 epochs and batch size of 1,024 were used to achieve convergence and acceptable accuracy of relative cell abundance estimation.
6. Estimating relative cell abundance in spatial data: learning rate of 0.01 (default), 20,000 epochs and batch size of 1,393 (full data) were used to achieve acceptable accuracy of relative cell abundance estimation.

Quantitative assessment of the specificity and sensitivity of cell2location at excluding cell types that should be absent in the spatial data. For all methods, precision recall was quantified for all 121 cell types and five Visium tissue sections jointly. All analyses and visualizations are based on the 99.8% quantile of relative cell abundance of each cell type in each Visium sample from the primary SSp cortex region as a predictor (Supplementary Fig. 11e), representing the presence of a given cell type at least at some locations within the sample. The cell count >3 in reference data from the SSp cortex region was used as a gold standard label, yielding $n = 98$ positive cell types and $n = 23$ negative cell types (Supplementary Fig. 11e). Identical criteria were used for alternative methods (Fig. 2i,j and Supplementary Fig. 11f), after the estimation of relative cell abundances for individual locations using those methods (see above).

Quantitative assessment of the consistency of the estimated relative cell abundance to reference data from manually dissected cortical layers. Cell2location cell type abundance estimates were averaged and normalized per cortical layer to obtain the proportion of cells per cell type within each cortical layer. These estimates were then compared to the cell proportions in the reference scRNA data (Supplementary Fig. 11a–d), considering 23 most prevalent cell types exclusive to the SSp cortex (cell count in SSp cortex >15 and cell count = 0 in the hippocampal region in the scRNA data). The same criteria were used to assess alternative methods (Supplementary Fig. 11d; RCTD, Stereoscope, SPOTlight, Seurat V3 and NNLS) after the estimation of relative cell abundance as described above.

Analysis of human lymph node Visium data. Construction of single-cell reference data. Published scRNA-seq datasets of lymph nodes have typically lacked an adequate representation of GC-associated immune cell populations due to the age of patient donors^{62,63}. We, therefore, included scRNA-seq datasets spanning lymph nodes, spleen and tonsils in our single-cell reference to ensure that we captured the full diversity of immune cell states likely to exist in the spatial transcriptomic dataset (Fig. 4a,b). The scRNA-seq reference was constructed by combining data from three studies^{35–37}. Integration was performed using scanpy⁵⁹, including removal of doublets with scrublet⁵⁸, regressing out batch from principal component space and performing BBKNN integration. The integrated reference was used to harmonize cell type and subtype labels among datasets, yielding 34 annotated cell populations represented across 73,260 cells.

Spatial data. 10x Visium spatial transcriptomics data of human lymph nodes were downloaded from the 10x Genomics website. A paired histology image was used to annotate GC locations.

Spatial mapping using cell2location. Expression signatures of 34 cell populations (Fig. 4b) were estimated using the negative binomial regression model (Supplementary Methods), accounting for the effect of different batches as well as technologies (Supplementary Fig. 24). Before model-based estimation, an untransformed and unnormalized count matrix was filtered to select expressed genes at two cutoffs:

1. Selecting genes detected at mRNA count >0 in >3% of cells
2. Selecting genes detected at mRNA count >0 in a few cells (3% > cell count > 0.1%) but with a mean expression across non-zero cells >1.122018

The untransformed and unnormalized mRNA counts from a single Visium experiment were used as input to cell2location using the hyper-parameters stated in Table 1. The prior cell abundance $\tilde{N} = 30$ was derived by examining histology images of lymph node tissue (paired images had insufficient image quality).

NMF. Absolute cell type abundance estimates obtained from cell2location were used as input for NMF to identify spatially interlaced tissue zones/compartments of lymph nodes (Fig. 4d). In brief, NMF decomposes cell abundance estimates w_{sf} into additive components:

$$w_{sf} = \sum_r z_{s,r} x_{r,f}$$

where $x_{r,f}$ represents NMF weight describing the contribution of each cell type group r to the abundance of cell types f , and $z_{s,r}$ represents NMF weight describing the abundances of each cell type group r across locations s .

NMF was trained for a range of $R = \{8, \dots, 28\}$, and the decomposition into 14 factors was chosen as a balance between capturing fine tissue zones and splitting known compartments such as light zone of the GC into several distinct factors (Fig. 4e, panel 3) into separate zones. NMF weight $x_{r,f}$ is shown in Fig. 4d and Supplementary Fig. 20c.

Cell2location estimates from this section are reported in Fig. 4, Supplementary Figs. 12b and 20–24 and Supplementary Files 10 and 11.

Assessment of cell2location and alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS) were used with default parameters as described in the section on synthetic data (Methods). Stereoscope (implementation using scvi-tools; see Methods, synthetic data section) was used with the following hyper-parameters:

7. No HVG selection was used (10,230 genes shared between single-cell and spatial data).
8. Estimating expression signatures of cell types: learning rate of 0.01 (default); 1,000 epochs (increased compared to default to achieve convergence); batch size of 256 (default)
9. Estimating relative cell abundance in spatial data: learning rate of 0.01 (default); 20,000 epochs (increased compared to default to achieve convergence); batch size of 256 (default)

Assessment criteria of cell2location and alternative methods. The accuracy of cell2location and alternative methods was assessed by mapping six GC cell populations (B_Cycling, B_GC_DZ, B_GC_LZ, B_GC_prePB, FDC and T_CD4⁺_TH1_GC)³⁷ to the GC locations, which were anatomically annotated using histology images paired to the Visium data (Fig. 4f). Precision recall, false-positive rate and true-positive rate were evaluated as a macro-average and separately for each cell type (scikit-learn) across six GC cell types expected to be present in annotated GC locations (positive set of locations) and absent from other locations (negative set) (Fig. 4f–h and Supplementary Fig. 23). Precision recall curves are shown for every cell type (Fig. 4g and Supplementary Fig. 23). To simplify the presentation, we additionally used bar plots (Fig. 4h and Supplementary Fig. 22), where bar height represents average precision recall score across cell types, and error bars show standard error of the mean.

Six known GC cell types were classified as either transcriptionally fine or transcriptionally distinct, based on the Euclidean distance of their expression signatures (linear mRNA count space rather than log-transformed) to the closest cell type from the full set of 34 cell types. Euclidean distance represents both the number of genes marking each cell type and the absolute expression levels of those genes, both of which are important for mapping. Cell types were classified as transcriptionally distinct if the Euclidean distance was greater than 100 and as transcriptionally fine if it was less than 100 (Supplementary Fig. 22a).

Additionally, we performed the comparison of model-based and hard-coded methods (Supplementary Methods, sections 2 and 3) for estimating the reference signatures of cell types and other cell2location modeling choices (see Supplementary Fig. 24 and the modeling choices section above), keeping all other parameters constant.

Analysis of human gut Visium data. *Construction of single-cell reference data.* We used a published gut scRNA dataset that covers a range of time points and gut regions from Elmentaite et al.⁴². Crucially, the dataset contains the cells from lymphoid structures within the gut (pediatric subset in particular) and covers the age range of the considered spatial data (adult gut). Literature evidence was used to annotate 19 of 76 cell types as expected in the lymphoid follicle.

Spatial data. 10x Visium spatial data consisting of two replicates of the biopsy of colon tissue were obtained from Fawcner-Corbett et al.⁴³. Crucially, the dataset contains mature lymphoid follicles. Both the paired histology image and expression of marker genes (Supplementary Fig. 25b) were considered to annotate lymphoid follicles, prioritizing histology when marker genes were detected outside expected histology areas; the annotation was performed using the 10x Loupe browser. Replicates also contain distinct sequencing depth/RNA detection sensitivity, requiring normalization across batches.

Spatial mapping using cell2location. Reference signatures for 76 cell types were estimated using the negative binomial regression model, accounting for the effect of different batches and technologies (3' versus 5') with training mini-batch size of 2,500 and training duration of 200 epochs. Before model-based estimation, an untransformed and unnormalized count matrix was filtered to select expressed genes at two cutoffs:

1. Selecting genes detected at mRNA count >0 in >3% of cells
2. Selecting genes detected at mRNA count >0 in a few cells (3% > cell count > 0.02%) but with a mean expression across non-zero cells >1.12.

The untransformed and unnormalized mRNA counts from a single Visium experiment were used as input to cell2location using the hyper-parameters stated in Table 1. The prior cell abundance $\tilde{N} = 25$ was derived by computing average cell count in the paired histology images for 20 random locations.

Cell2location estimates from this section are reported in Fig. 5, Supplementary Figs. 25 and 26 and Supplementary Files 12 and 13.

Assessment of cell2location and alternative methods. RCTD, SPOTlight, Seurat 3 and NNLS (AutoGeneS) were used with default parameters as described in the section on synthetic data (Methods). Stereoscope (implementation using scvi-tools; see Methods, synthetic data section) was used with the following hyper-parameters:

10. No HVG selection was used (14,388 genes shared between single-cell and spatial data).
11. Estimating expression signatures of cell types: learning rate of 0.01 (default); 8,000 epochs (increased compared to default to achieve convergence); batch size of 1,024 (increased compared to default in an attempt to achieve sufficient accuracy)
12. Estimating relative cell abundance in spatial data: learning rate of 0.01 (default); 50,000 epochs (increased compared to default to achieve convergence); batch size of 1,024 (increased compared to default in an attempt to achieve sufficient accuracy)

Similarly to lymph node analysis described above, precision recall curves and scores were used to compare alternative methods. Cell proportions across locations for each of the 19 cell types expected in the lymphoid follicles were used as a predictor; histology-based annotation of the lymphoid follicles was used as a positive set; and all other locations were used as a negative set (Supplementary Figs. 25d and 26). Again, these 19 cell types were classified into transcriptionally fine and transcriptionally distinct using Euclidean distance to the closest other cell type as described in the lymph node section above (Fig. 5f and Supplementary Fig. 25d).

Importance of model features (Supplementary Fig. 25e) and sensitivity to hyper-parameters (Supplementary Fig. 5b) were assessed using precision recall procedures as described above and in the section about sensitivity of model features and hyper-parameters.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

Data generated for this manuscript (snRNA-seq and Visium from adjacent sections in the mouse brain (Fig. 2a), sequencing reads as well as Cell Ranger and Space Ranger output) were submitted to ArrayExpress under accession numbers E-MTAB-11114 (Visium) and E-MTAB-11115 (snRNA-seq). Annotated snRNA-seq data are publicly available via cellxgene portals⁶⁴: full dataset (annotation_1_print column denotes cell types) and astrocyte subclusters. The integrated secondary lymphoid organ scRNA-seq data are publicly available for download through S3 bucket. Image data generated in this manuscript were deposited to the BioImage Archive (accession no. S-BIAD207). Ground truth annotations of germinal center zones in lymph node Visium data can be found on GitHub. Ground truth annotations of the gut lymphoid follicles can be found on GitHub. Published datasets. snRNA-seq data from Yao et al. were downloaded from the Allen Brain Institute data portal. Slide-seq V2 data from Stickels et al. were downloaded from the Broad Institute data portal (subject to use agreement). Visium data of human lymph nodes can be downloaded from the 10x website (via a function in the scanpy package).

Code availability

The cell2location package is available at <https://github.com/BayraktarLab/cell2location/>. Documentation and tutorials are available at <https://cell2location.readthedocs.io/>. Code used to generate synthetic data is available at https://github.com/vitkl/cell2location_paper/blob/master/notebooks/benchmarking/synthetic_data_construction_improved_real_mg.ipynb. Code used to segment nuclei in histology images is available at <https://github.com/yozhikoff/segmentation>. Jupyter notebooks covering the analysis in this paper are available at https://github.com/vitkl/cell2location_paper/ and upon reasonable request.

References

56. Hafemeister, C. & Satija, R. Normalization and variance stabilization of single-cell RNA-seq data using regularized negative binomial regression. *Genome Biol.* **20**, 296 (2019).
57. Krishnaswami, S. R. et al. Using single nuclei for RNA-seq to capture the transcriptome of postmortem neurons. *Nat. Protoc.* **11**, 499–524 (2016).
58. Wolock, S. L., Lopez, R. & Klein, A. M. Scrublet: computational identification of cell doublets in single-cell transcriptomic data. *Cell Syst.* **8**, 281–291 (2019).
59. Wolf, F. A., Angerer, P. & Theis, F. J. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol.* **19**, 15 (2018).
60. Roberts, K. & Bayraktar, O. A. Automation of multiplexed RNAscope single-molecule fluorescent in situ hybridization and immunohistochemistry for spatial tissue mapping. *Methods Mol. Biol.* **2148**, 229–244 (2020).
61. Berg, S. et al. ilastik: interactive machine learning for (bio)image analysis. *Nat. Methods* **16**, 1226–1232 (2019).
62. Turner, J. S. et al. Human germinal centres engage memory and naive B cells after influenza vaccination. *Nature* **586**, 127–132 (2020).
63. Madissoon, E. et al. scRNA-seq assessment of the human lung, spleen, and esophagus tissue stability after cold preservation. *Genome Biol.* **21**, 1 (2019).
64. Megill, C. et al. chanzuckerberg/cellxgene: release 0.15.0. <https://doi.org/10.5281/zenodo.3710410> (2020).

Acknowledgements

We thank B. Velten, Y. Huang and L. Marconato for feedback on the cell2location model; M. Prete and V. Kiselev for dockerizing the cell2location tool and creating the web portals for sharing our data; N. Kumasaka for helpful comments on single-cell analysis; S. Leonard and K. Polanski for help with spatial and single-nucleus data processing; K. James for advice on gut immune cell types; K. Roberts for advice on smFISH; J. E. Kwa for advice on snRNA-seq; J. Eliasova for illustrations and logo design; K. James advice on human gut immune cells; A. Antanaviciute, H. Koohy and A. Simmons for sharing gut Visium data; F. Obermeyer and M. Jankowiak for help with advanced use of pyro code base; and D. Rowitch and S. Teichmann for comments on the manuscript. H.W.K. was funded by a Sir Henry Wellcome Postdoctoral Fellowship (213555/Z/18/Z). This study was supported by Wellcome Trust Core Funding to O.A.B.

Author contributions

V.K., O.S. and O.A.B. conceived the study. V.K. developed and implemented the cell2location model with feedback from O.S. and worked on validation, mouse brain, human lymph node, gut and benchmarking analyses. A.S. performed Slide-seq and hyperparameter sensitivity analysis, performed histology image segmentation, wrote data visualization modules and contributed to implementation of cell2location in pyro and benchmarking analyses. E.D. worked on generating synthetic data, mouse brain cell annotation and validation of cell type mapping and contributed to lymph node analysis. V.K.E. and A.S. contributed to the development and interpretation of the model for estimating cell type signatures. A.L., M.G., A.A.I. and M.S.J. contributed to the developing of model architecture, writing the model and downstream analysis code, interpretation of the cell2location results as well as downstream clustering and NMF. A.A.R. performed benchmarking of stereoscope, supervised by R.V.-T. and O.S. A.G. contributed to implementation of cell2location using scvi-tools and pyro. H.W.K. contributed to lymph node data analysis and interpretation, supervised by L.J. R.E. contributed to gut data analysis. L.R. performed snRNA-seq experiments. L.T. performed Visium experiments. J.S.P. performed smFISH experiments and imaging. T.L. performed smFISH image processing and astrocyte segmentation analysis. V.K., O.S. and O.A.B. wrote the manuscript, with feedback from all authors.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41587-021-01139-4>.

Correspondence and requests for materials should be addressed to Oliver Stegle or Omer Ali Bayraktar.

Peer review information *Nature Biotechnology* thanks Steven Sloan, Xiaoqun Wang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Reporting Summary

Nature Research wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Research policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☒ ☐ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☒ ☐ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☐ ☒ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection 10X CellRanger version 3.0.2, 10X SpaceRanger version 1.0

Data analysis The cell2location package is available at <https://github.com/BayraktarLab/cell2location/>. Documentation and tutorials are available at <https://cell2location.readthedocs.io/>. Code used to generate synthetic data is available in https://github.com/vitkl/cell2location_paper/blob/master/notebooks/benchmarking/synthetic_data_construction_improved_real_mg.ipynb. Code used to segment nuclei in histology images is available in <https://github.com/yozhikoff/segmentation>. Jupyter notebooks covering the analysis in this paper are available at https://github.com/vitkl/cell2location_paper and upon request.

Specific package version used for analysis are as follows (python):

```
python 3.7.10
scvi 0.13.0
pyro 1.6.0+823dce2b
scanpy 1.7.2
numpy 1.20.2
scipy 1.6.2
sklearn 0.24.1
Other python dependencies:
torch 1.8.1+cu102
anndata 0.7.5
pandas 1.2.3
leidenalg 0.8.3
louvain 0.7.0
```

Specific package version used for analysis are as follows (R):

R 4.0.3

Seurat 4.0

SeuratDisk 0.0.0.9019

RCTD (version from github commit 037a5da2c282fab94b1aee5cf489de73c494dc39)

R SPOTlight was used via the docker image provided by the developers (marcelosua/spotlight_env_r:latest):

SPOTlight 0.1.7

R 3.6.3

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Data generated for this manuscript (snRNA-seq and Visium from adjacent sections in the mouse brain, Fig 2A, sequencing reads as well as CellRanger and SpaceRanger output) is submitted to ArrayExpress under accession numbers E-MTAB-11114 (Visium) and E-MTAB-11115 (snRNA-seq). Image data generated in this manuscript was deposited to BioImage Archive (accession number: S-BIAD207).

The integrated secondary lymphoid organ scRNA-seq data are publicly available for download through S3 buckets at <https://cell2location.cog.sanger.ac.uk/browser.html> (lymph nodes - /paper/integrated_lymphoid_organ_scrna/). snRNAseq data can be publicly viewed without download via cellxgene portals at <https://cell2location.cellgeni.sanger.ac.uk/mouse-brain> (full dataset, annotation_1_print column denotes cell types) and <https://cell2location.cellgeni.sanger.ac.uk/mouse-brain-astrocytes> (astrocyte subclusters).

Ground truth annotations of Germinal Center zones in lymph node Visium data can be found at https://github.com/vitkl/cell2location_paper/blob/master/notebooks/selected_results/lymph_nodes_analysis/manual_GC_annot.csv. Ground truth annotations of the gut lymphoid follicles can be found at https://github.com/vitkl/cell2location_paper/tree/master/notebooks/selected_results/tissue_annotation/oxford.

Published datasets. snRNA-seq data from Yao et al were downloaded from Allen Brain Institute data portal (<https://portal.brain-map.org/atlas-and-data/rnaseq/mouse-whole-cortex-and-hippocampus-smart-seq>). Slide-Seq V2 data from Stickels, Murray et al were downloaded from the Broad Institute data portal (https://singlecell.broadinstitute.org/single_cell/study/SCP815/sensitive-spatial-genome-wide-expression-profiling-at-cellular-resolution#study-download, subject to use agreement). Visium data of human lymph nodes can be downloaded from 10X website (via a function in Scanpy package).

Genome references used for mapping mouse brain data were downloaded from 10X website.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

We aimed to profile 2 mice at 4 adjacent tissue positions (i.e. sections spaced by 200 microns). In total, four consecutive sets of thick and thin tissue sections were collected from each brain. Five sets of tissue sections yielded both good quality single nuclei and Visium data (three adjacent sections from mouse 1 and two sections from mouse 2) while one additional section from mouse 2 yielded good single nuclei; these were sequenced and considered for analysis in this study (Fig 2-3).

For analysis of other tissues and Slide-Seq V2 published spatial transcriptomics datasets were used (human lymph node, n=1; human gut, n=2; Slide-Seq V2, n=1) with the total number of samples limited by availability.

No statistical methods were used to pre-determine sample sizes but our sample sizes are similar to those reported previously (references 3, 25, 42-43).

Data exclusions

Visium spots located outside tissue areas were excluded from all analyses in the paper based on the histological examination. Specifically this was done in 2 steps:

1. Automatic tissue detection using SpaceRanger workflow with subsequence location filtering. No further data exclusion for mouse brain and human lymph node data were performed.
2. When the automatic tissue detection failed (adult human gut samples) - manual annotation was performed, with subsequent filtering of location that do not overlap tissue.

The rationale behind this filtering is as follows: 10X Visium locations are manufactured in a hexagonal grid within square capture areas - as a

	result not all locations are covered by tissue and the locations that are not covered contain background RNA rather than RNA from cells in the tissue.
Replication	For mouse experiments, multiple animals were processed independently on the same day. Fig 2G and Fig S10 report on consistency of cell2location cell abundance estimates between adjacent sections and replicate animals. Differences cell type abundances between sections can be attributed to change in the regional composition (by histology examination). Cell type locations of novel astrocyte subtypes were validated using single molecule FISH of mRNA for gene marking the novel subtypes at corresponding anatomical location in different mice (Fig 3D) - thus replicating the detection of novel subtypes using alternative technology and an independently processed animal.
Randomization	Randomization is not relevant to our study because only wild-type mice were used for mouse brain mapping experiments. For lymph node and gut mapping experiments, the data were derived from published observational studies.
Blinding	Blinding is not relevant to our study because the objective was not to compare the effect of 2 conditions. Real world data were generated from the mouse brain or derived from published observational studies that included wild-type / healthy tissues only and the simulations were generated by the research team.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Animals and other organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Laboratory animals	Two wild-type adult C57BL/6 mice (postnatal day 56, 1 female and 1 male) were used for 10X Visium experiments. One additional wild-type adult C57BL/6 mouse (postnatal day 63, male) was used for single-molecule fluorescent in situ hybridization.
Wild animals	Study did not involve wild animals.
Field-collected samples	Study did not involve field-collected samples.
Ethics oversight	Mouse work was carried out at the Wellcome Sanger Institute in accordance with UK Home Office regulations and UK Animals (Scientific Procedures) Act of 1986 under a UK Home Office license, which were regularly reviewed by the institutional Animal Welfare and Ethical Review Body.

Note that full information on the approval of the study protocol must also be provided in the manuscript.